

Modeling Arteriolar Flow and Mass Transport Using the Immersed Boundary Method

Kayne M. Arthurs,^{*} Leon C. Moore,[†] Charles S. Peskin,[‡]
E. Bruce Pitman,[§] and H. E. Layton^{*}

^{}Department of Mathematics, Duke University, Durham, North Carolina 27708-0320; [†]Department of Physiology and Biophysics, State University of New York, Stony Brook, New York 11794-8661; [‡]Courant Institute of Mathematical Sciences, New York University, New York, New York 10012; [§]Department of Mathematics, State University of New York, Buffalo, New York 14214-3093*

E-mail: ksmith@math.duke.edu, moore@physiology.pnb.sunysb.edu, peskin@cims.nyu.edu, pitman@sand.math.buffalo.edu, layton@math.duke.edu

Received April 28, 1998; revised August 14, 1998

Flow in arterioles is determined by a number of interacting factors, including perfusion pressure, neural stimulation, vasoactive substances, the intrinsic contractility of arteriolar walls, and wall shear stress. We have developed a two-dimensional model of arteriolar fluid flow and mass transport. The model includes a phenomenological representation of the myogenic response of the arteriolar wall, in which an increase in perfusion pressure stimulates vasoconstriction. The model also includes the release, advection, diffusion, degradation, and dilatory action of nitric oxide (NO), a potent, but short-lived, vasodilatory agent. Parameters for the model were taken primarily from the experimental literature of the rat renal afferent arteriole. Solutions to the incompressible Navier-Stokes equations were approximated by means of a splitting that used upwind differencing for the inertial term and a spectral method for the viscous term and incompressibility condition. The immersed boundary method was used to include the forces arising from the arteriolar walls. The advection of NO was computed by means of a high-order flux-corrected transport scheme; the diffusion of NO was computed by a spectral solver. Simulations demonstrated the efficacy of the numerical methods employed, and grid refinement studies confirmed anticipated first-order temporal convergence and demonstrated second-order spatial convergence in key quantities. By providing information about the effective width of the immersed boundary and shear stress magnitude near that boundary, the grid refinement studies indicate the degree of spatial refinement required for quantitatively reliable simulations. Owing to the dominating effect of NO advection, relative to degradation and diffusion, simulations indicate that NO has the capacity to produce dilation along the entire length of the arteriole. © 1998 Academic Press

Key Words: computational biofluidynamics; blood vessel; renal microvasculature; myogenic autoregulation; nitric oxide.

1. INTRODUCTION

The goals of this study are to formulate a detailed model for blood flow and mass transport in an arteriole and to develop and evaluate a numerical algorithm for the solution of the model. The model and numerical algorithm rely heavily on the immersed boundary method (IMBM), which was originally developed to study fluid motion in the heart [27].

Arterioles have diameters ranging from about 10 to 200 μm and lengths ranging from 100 to 1000 μm . Average flow velocities are ~ 1 cm/s, and pressure gradients along the direction of flow are ~ 1000 mm Hg/cm. The Reynolds number of arteriolar flow is typically about 0.1 (based on an arteriolar diameter of 20 μm and a kinematic blood viscosity of 1.66×10^{-2} cm^2/s [1, 23]).

Arteriolar walls consist of several outer layers of smooth muscle cells and an inner layer of endothelial cells. These endothelial cells can release vasoactive substances into the blood stream and into the arteriolar wall, where they can influence the adjacent smooth muscle cells. In this study, the arteriolar wall is modeled as a massless, elastic–contractile closed boundary immersed in a viscous, incompressible fluid. The boundary is initially configured as parallel line segments with semi-circular closures at the ends (see Fig. 1). The fluid region inside the boundary represents blood; the exterior fluid region represents the local surroundings of the arteriole. Flow through the model arteriole is induced by a high-pressure fluid source at one end and by a low-pressure fluid sink at the opposite end.

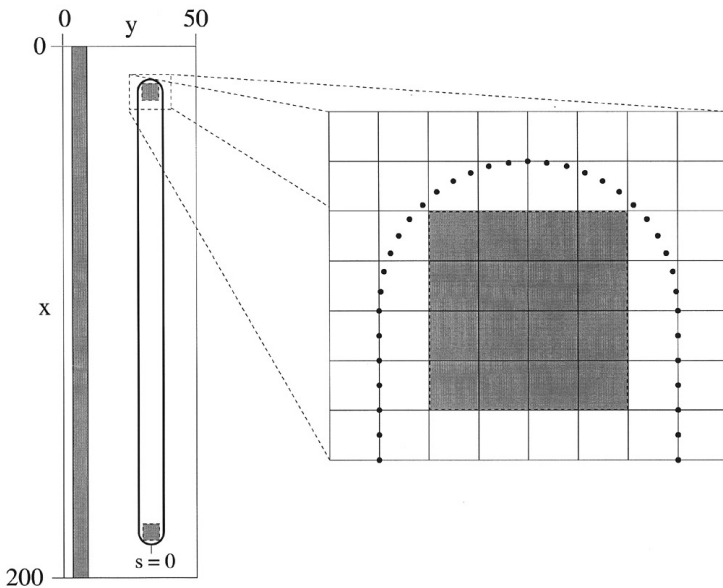


FIG. 1. The domain Ω , with model configuration, is shown at left. The initial position of the model arteriolar wall (dark heavy line) encloses the fluid source (upper shaded square) and sink (lower shaded square), which have fixed positions. The shaded vertical strip is the support of the external source/sink. An enlarged portion of Ω , at right, shows the support of the source, relative to the Eulerian grid, and the individual nodes used in the numerical discretization of the arteriolar wall (see Section 3). Length scales are in micrometers.

Fluid velocities and pressures are determined by solving the incompressible Navier–Stokes equations, modified in accordance with the IMBM (see below). After pressurization, the model arteriole is approximately $20\ \mu\text{m}$ in diameter and $150\ \mu\text{m}$ in length.

The description of the boundary includes a phenomenological model of the myogenic response, in which an increase in intravascular pressure causes a contraction of vascular smooth muscle. This contraction reduces average arteriolar diameter, which increases total segmental resistance and thereby acts to stabilize downstream pressure [5]. In vivo, this contractile response is modulated by the local concentration of vasoactive agents, e.g., nitric oxide (NO), a potent vasodilator. The release of NO from the arteriolar wall is stimulated by shear stress [4] and by the binding of a number of other vasoactive substances [24]. Alternatively, NO may be released as a consequence of vascular injury [18]. The arteriolar model includes provisions for the mass transport of NO, specifically, the release, advection, diffusion, degradation, and vasodilatory action of NO. Mathematical models have been developed by other investigators to examine the temporal evolution of NO in tissue [8, 22]. Our model is distinguished by its emphasis on the detailed description of fluid motion and coupled NO transport.

The IMBM was used to model the interaction between the fluid and the arteriolar wall by means of an immersed elastic–contractile boundary. The action of the boundary on the fluid arises from a body force included in the Navier–Stokes equations. The boundary, as a consequence of the no-slip condition, is required to move at the local fluid velocity. The IMBM has been used to simulate fluid dynamics of the inner ear [2], bacterial swimming [9], biofilm processes [10], peristaltic pumping of solid particles [11], sperm motility in the presence of boundaries [12], aquatic animal locomotion [13], platelet aggregation during blood clotting [14], suspended particles in three-dimensional Stokes flow [15], and three-dimensional heart motion [29].

The Navier–Stokes equations were solved by a tripartite splitting that is first-order in time. The body force arising from the boundary was evaluated by an explicit differencing. The inertial term, which makes only a small contribution to the solution for low Reynolds number flow, was evaluated by upwind differencing. The viscous term and the incompressibility condition were considered together in a backward Euler discretization of the time-dependent Stokes equations. These equations were then evaluated by a spectral method.

The advection–diffusion equation for mass transport was also solved by a tripartite splitting. Solute sources were evaluated by an explicit differencing. The advection term, which requires special care to avoid artifactual diffusion and dispersion, was evaluated by a flux-corrected transport scheme that employs upwind-differencing for the low-order component and an eighth-order Richtmyer two-step Lax–Wendroff scheme for the high-order component. The diffusion term, represented by a backward Euler discretization, was evaluated by a spectral method.

Simulation parameters are based on experimental measurements from the renal afferent arteriole of the rat. The afferent arteriole has been heavily studied because of its role in hemodynamic control, which is critical to the homeostatic function of the kidney and which protects delicate down-stream structures (e.g., the glomerulus) [32].

Extensive tests were conducted to confirm the accuracy and adequacy of the numerical methods and to determine the degree of numerical resolution required for quantitatively reliable simulations. In particular, tests were conducted to determine the effective immersed boundary width and the shear stress magnitude near the boundary. Simulations indicate that the model exhibits first-order convergence in time and that key quantities exhibit

second-order convergence in space. Simulations also indicate that a spatial grid resolution of $<1 \mu\text{m}$ is required to adequately represent mass transport of NO.

2. MODEL FORMULATION

2.1. Fluid Equations and Immersed Boundary Method

In this two-dimensional model, the arteriolar wall is modeled as an immersed boundary in a rectangular Eulerian fluid domain Ω with periodic boundary conditions. The domain Ω , expressed in units of μm , is the set $[0, 200] \times [0, 50]$; position in Ω will be denoted by $\mathbf{x} = (x, y)$, with the positive x -direction usually pointing downward in our figures (see Fig. 1). The fluid is assumed to be incompressible with uniform density ρ and uniform dynamic viscosity μ . The immersed boundary is initially situated within Ω as shown in Fig. 1. The boundary configuration at time t is given parametrically by $\mathbf{X}(s, t)$, $-l/2 \leq s \leq l/2$, where $\mathbf{X}$ is a point in Ω , the parameter s marks a fixed material point of the boundary, and l is the length of the unstressed boundary. Because the boundary is closed, $\mathbf{X}(-l/2, t) = \mathbf{X}(l/2, t)$. The value $s = 0$ corresponds to the point on the boundary having the largest x -coordinate. The parameter s increases with counterclockwise orientation.

The Navier–Stokes equations, modified for our application, are given by

$$\rho(\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u}) = -\nabla p + \mu \Delta \mathbf{u} + \mathbf{F} - \sigma \kappa \mathbf{u}, \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0, \quad (2)$$

where $\mathbf{u}(\mathbf{x}, t)$ is the fluid velocity and $p(\mathbf{x}, t)$ is the fluid pressure. The body forces that the immersed boundary imparts to the fluid are included as an external force density $\mathbf{F}(\mathbf{x}, t)$. To simulate the anatomical structures and interstitial matrix external to the arteriole, a damping term, $\sigma \kappa \mathbf{u}$, proportional to the fluid velocity and with constant damping coefficient κ has been added to the Navier–Stokes equations. The dimensionless function $\sigma(\mathbf{x}, t)$ is constructed so that damping acts externally to the vessel; σ is approximately one outside the model vessel and zero inside the vessel (see Eq. (31) below). The damping term, by reducing the velocity of flow that is induced exterior to the model vessel (by expansion of the vessel), allows more rapid attainment of a near steady state. The divergence-free condition given by Eq. (2) applies over Ω except at fluid sources or sinks (see below).

The force density $\mathbf{F}(\mathbf{x}, t)$, a Dirac delta-function layer along the immersed boundary, is given by

$$\mathbf{F}(\mathbf{x}, t) = \int_{-l/2}^{l/2} \mathbf{f}(s, t) \delta(\mathbf{x} - \mathbf{X}(s, t)) ds, \quad (3)$$

where δ is the two-dimensional Dirac delta function and $\mathbf{f}(s, t)$ is the density of the boundary force with respect to the measure ds . The boundary force density $\mathbf{f}(s, t)$ arises from the properties of the boundary and, therefore, is specific to the particular application of the method. To represent arteriolar walls, $\mathbf{f}(s, t)$ should include the elastic–contractile properties of the smooth muscle sheath and its modulation by vasoactive agents.

Because the immersed boundary moves with the fluid, the equation of motion for the boundary, equivalent to the no-slip condition, is given by

$$\frac{\partial \mathbf{X}(s, t)}{\partial t} = \mathbf{u}(\mathbf{X}(s, t), t) = \int_{\Omega} \mathbf{u}(\mathbf{x}, t) \delta(\mathbf{x} - \mathbf{X}(s, t)) d\mathbf{x}. \quad (4)$$

The relationships set forth in Eqs. (3) and (4), which describe coupling between the boundary and the fluid, are key features of the IMBM.

To provide for fluid flow through the lumen of the model arteriole, a fluid source is fixed at the upper end of the model vessel and a sink is fixed at the lower end; the support of the source and sink are illustrated in Fig. 1 as shaded squares. An external source/sink, located exterior to the model vessel, ensures the conservation of fluid in Ω ; this external source/sink is illustrated in Fig. 1 as a shaded strip parallel to the x -axis.

The locations and strengths of the external source/sink, the vessel source, and the vessel sink are specified by weighting functions $w_m(\mathbf{x})$, with $m = 0, 1, 2$, respectively. Each function w_m is nonnegative and has weight one: $\int_{\Omega} w_m(\mathbf{x}) d\mathbf{x} = 1$. The functions w_1 and w_2 are centered at the locations $\mathbf{x}_{S1} = (18.75, 31.25)$ and $\mathbf{x}_{S2} = (181.25, 31.25)$, respectively, and they have support contained in squares of side $6.25 \mu\text{m}$. The external source/sink w_0 has support in the rectangle $[0, 200] \times [3.125, 9.375]$, which has centerline $y_{S0} = 6.25 \mu\text{m}$. This centerline and the line $y = 31.25$ through the vessel source and sink are lines of symmetry in Ω .

The rate of flow from source m that emerges from a point in Ω is the product of the weight $w_m(\mathbf{x})$ and the source flow rate, given by $q_m(t)$. Since the external source/sink conserves fluid volume,

$$q_0(t) = - \sum_{m=1}^2 q_m(t). \tag{5}$$

The incompressibility condition, Eq. (2), modified to include flow from sources and sinks and taking Eq. (5) into account, is given by

$$\nabla \cdot \mathbf{u} = \sum_{m=1}^2 q_m(t)(w_m(\mathbf{x}) - w_0(\mathbf{x})). \tag{6}$$

Note that

$$\int_{\Omega} \nabla \cdot \mathbf{u} d\mathbf{x} = \sum_{m=1}^2 q_m(t) \int_{\Omega} (w_m(\mathbf{x}) - w_0(\mathbf{x})) d\mathbf{x} = 0. \tag{7}$$

If this were not true, we would have a contradiction, because $\mathbf{u}$ is assumed to be a smooth periodic vector field, and by the divergence theorem, the integral of $\nabla \cdot \mathbf{u}$ over its period is zero.

The flow rates $q_1(t)$ and $q_2(t)$ are determined by assuming that short vessel segments, not in Ω , are attached to the vessel's source and sink. The pressure $\bar{P}_1$ upstream from the source and the pressure $\bar{P}_2$ downstream from the sink are specified as parameters, along with the resistances R_1 and R_2 in the corresponding segments. If one assumes that flow in the short segments is approximated by Poiseuille flow, the flow rates $q_1(t)$ and $q_2(t)$ are given by the Hagen–Poiseuille equation,

$$q_m(t) = \frac{\bar{P}_m - P_m(t)}{R_m}, \tag{8}$$

where $P_m(t)$ is the weighted average of the pressure over the source (or sink), scaled relative to the pressure over the external source/sink, i.e.,

$$P_m(t) = \int_{\Omega} p(\mathbf{x}, t)(w_m(\mathbf{x}) - w_0(\mathbf{x})) d\mathbf{x}, \quad m = 1, 2. \tag{9}$$

Once q_1 and q_2 are computed, q_0 follows from Eq. (5). Although in this study the external pressures $\bar{P}_m$ and resistances R_m are defined to be constant, one may allow them to vary in time.

2.2. Force Density Equations

The force density exerted by the immersed boundary arises from the properties of the boundary. For the model arteriole, the boundary force density $\mathbf{f}(s, t)$ has three components:

- (i) an elastic component $\mathbf{f}_B(s, t)$ generated by tension in the boundary;
- (ii) a tether component $\mathbf{f}_T(s, t)$ that arises from elastic connections between the boundary and specific points in Ω ; and
- (iii) an elastic–contractile component $\mathbf{f}_M(s, t)$, which is typically nearly perpendicular to the boundary and which depends on the local intravascular pressure and the local concentrations of vasoactive substances.

The component of force density generated by tension in the boundary is given by

$$\mathbf{f}_B = \frac{\partial(B\boldsymbol{\tau}_B)}{\partial s}, \quad (10)$$

where the tension B , which arises from boundary strain at all points along the boundary, is given by a continuous version of Hooke's law with spring constant K_B :

$$B(s, t) = K_B \left(\left| \frac{\partial \mathbf{X}(s, t)}{\partial s} \right| - 1 \right). \quad (11)$$

The direction of B is given by the unit tangent vector

$$\boldsymbol{\tau}_B(s, t) = \left(\frac{\partial \mathbf{X}(s, t)}{\partial s} \right) / \left| \frac{\partial \mathbf{X}(s, t)}{\partial s} \right|. \quad (12)$$

Tethers are used to stabilize the model arteriole in Ω , much as surrounding structures and connective tissue stabilize arterioles in vivo. In the absence of tethers, the expansion of the model vessel, upon pressurization, excites currents that cause the vessel to translate in the fluid domain. Since the formulation and solution of the model is simpler with sources that remain fixed in Ω , tethers serve the practical purpose of keeping the vessel nearly translation-free.

A tether is constructed as an elastic component connecting a material point s' on the boundary to a fixed anchor point $\mathbf{x}_T$ in Ω ; the elastic component may be allowed to expand to a given length L_T before a restoring force is activated. The force density arising from a single tether is given by

$$\mathbf{f}_T(s, t) = K_T \max\{0, (|\mathbf{x}_T - \mathbf{X}(s, t)| - L_T)\} \delta(s - s') \boldsymbol{\tau}_T(s, t), \quad (13)$$

where the unit vector $\boldsymbol{\tau}_T$ is given by

$$\boldsymbol{\tau}_T(s, t) = \begin{cases} \frac{\mathbf{x}_T - \mathbf{X}(s, t)}{|\mathbf{x}_T - \mathbf{X}(s, t)|}, & \mathbf{x}_T \neq \mathbf{X}(s, t), \\ 0, & \mathbf{x}_T = \mathbf{X}(s, t). \end{cases} \quad (14)$$

TABLE I
Initial Coordinates of Tether Attachments to Nodes (μm)

Tether	x	y	s
T1	10.937500000000	31.250000000000	$\pm l/2$
T2	10.977602212310	30.638158473969	$-l/2 + l_1/12$
T3	10.977602212310	31.861841526031	$+l/2 - l_1/12$
T4	11.097222689270	30.036785726082	$-l/2 + l_1/6$
T5	11.097222689270	32.463214273918	$+l/2 - l_1/6$
T6	15.625000000000	26.562500000000	$-l_2 - l_1$
T7	15.625000000000	35.937500000000	$+l_2 + l_1$
T8	184.375000000000	26.562500000000	$-l_1$
T9	184.375000000000	35.937500000000	$+l_1$
T10	189.062500000000	31.250000000000	0

Note. Here, $l = 4l_1 + 2l_2$, with $l_1 = (600/64)(\pi/4)$ and $l_2 = 108(100/64) = 168.75$. The total length of the unstressed boundary is given by l ; $2l_1$ is the length of the upper or lower unstressed semi-circle; l_2 is the unstressed length of the straight portion of a side. Each Lagrangian attachment point s remains fixed in time, but its Eulerian coordinate will evolve with the motion of the boundary.

The Dirac delta function in Eq. (13), with support at s' , specifies that the tether force acts at s' .

Ten tethers of the type that have just been described are connected to the immersed boundary that forms the model arteriole wall; their points of attachment to the boundary are given in Table I, and their anchor points are given in Table II. Tether placement is shown in Fig. 2. Five tethers with zero rest length ($L_T = 0$), labeled T1–T5, anchor the model arteriole above the fluid source. Four additional tethers, labeled T6–T9, are attached to the ends of the initially straight sections of the boundary, and a tenth, T10, is attached to the base of the boundary. For these tethers, $L_T = 200/256 \mu\text{m}$, which allows the model arteriole to expand and lengthen before a stabilizing force is applied.

The elastic–contractile component of force density, given by

$$\mathbf{f}_M(s, t) = M(s, t) \boldsymbol{\tau}_M(s, t), \tag{15}$$

TABLE II
Tether Anchor Locations (μm)

Tether	x	y
T1	10.937500000000	31.250000000000
T2	10.977602212310	30.638158473969
T3	10.977602212310	31.861841526031
T4	11.097222689270	30.036785726082
T5	11.097222689270	32.463214273918
T6	15.625000000000	25.781250000000
T7	15.625000000000	36.718750000000
T8	184.375000000000	25.781250000000
T9	184.375000000000	36.718750000000
T10	189.843750000000	31.250000000000

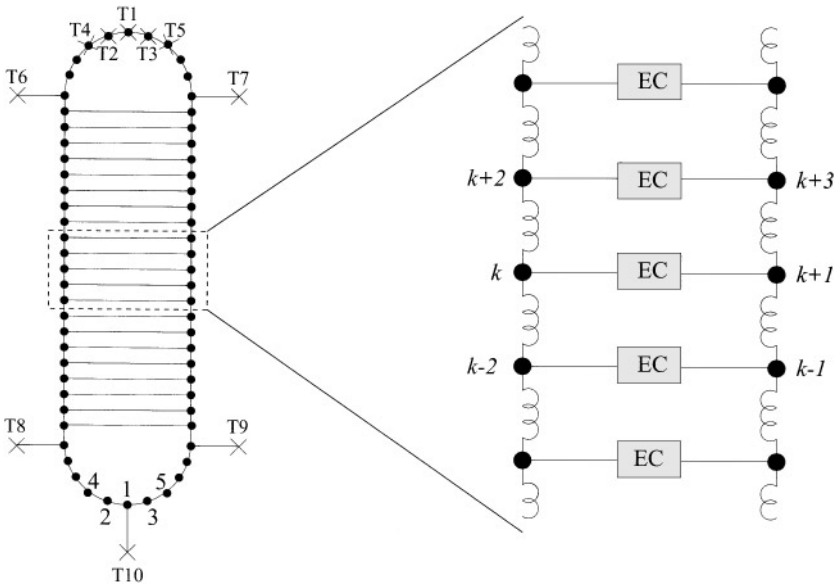


FIG. 2. At left is a schematic representation of the discretized boundary, with a reduced number of nodes. The lines between nodes indicate node connectivity. Tethers connect some nodes to fixed anchor points on Ω ; anchor points are indicated by an $\times$. The enlargement at right provides a schematic representation of the elastic-contractile elements (EC) that connect opposing nodes and the purely elastic elements that connect adjacent nodes. Boundary nodes are indexed on the right side of the boundary by odd integers and on the left by even integers.

contains the myogenic response of the vessel and largely determines arteriolar diameter. The form of $M(s, t)$ depends on the particular type of arteriole, and later, in Section 4, a specific formulation is developed for the renal afferent arteriole. However, in general, the myogenic force density at a material point s on the boundary is assumed to depend on the intravascular pressure $P(s, t)$ at the centerline of the model arteriole and on the concentration of a vasoactive substance at the boundary. Since $\mathbf{X}(s, t)$ and $\mathbf{X}(-s, t)$ are symmetric points with respect to the vessel centerline, the centerline is given by $\mathbf{G}(s, t) = (\mathbf{X}(s, t) + \mathbf{X}(-s, t))/2$, and $P(s, t)$ is given by

$$P(s, t) = \int_{\Omega} p(\mathbf{x}, t) (\delta(\mathbf{x} - \mathbf{G}(s, t)) - w_0(\mathbf{x})) d\mathbf{x}. \quad (16)$$

This pressure, which is measured relative to the external source/sink pressure, is a good approximation to the transmural pressure since the pressure everywhere external to the model arteriole is nearly equal to the external source/sink pressure. If $c(\mathbf{x}, t)$ is the vasoactive substance concentration defined on Ω , then the concentration at a site s on the boundary is

$$c(\mathbf{X}(s, t), t) = \int_{\Omega} c(\mathbf{x}, t) \delta(\mathbf{x} - \mathbf{X}(s, t)) d\mathbf{x}. \quad (17)$$

The myogenic force is applied along the sides of the model arteriole between the attachment points of tethers T6 and T8 and tethers T7 and T9. The myogenic force acts in the direction of the model arteriole centerline, given by

$$\tau_M(s, t) = \frac{\mathbf{X}(-s, t) - \mathbf{X}(s, t)}{|\mathbf{X}(-s, t) - \mathbf{X}(s, t)|}. \quad (18)$$

The total force density at a general material point s along the boundary at time t is given by

$$\mathbf{f}(s, t) = \mathbf{f}_B(s, t) + \mathbf{f}_T(s, t) + \mathbf{f}_M(s, t). \quad (19)$$

2.3. Mass Transport Equations

The advection and diffusion of solute dissolved in the fluid is coupled to the fluid velocity $\mathbf{u}$. The advection–diffusion equation, including the effects of both fluid and solute sources [31], is given for a solute of concentration $c(\mathbf{x}, t)$ by

$$c_t + \nabla \cdot (\mathbf{u}c) = \nabla \cdot (D\nabla c) + Q_S, \quad (20)$$

where $D(\mathbf{x})$ is solute diffusivity and $Q_S(\mathbf{x}, t)$ represents solute sources. For this study

$$Q_S(\mathbf{x}, t) = s_o(\mathbf{x}, t) - a(\mathbf{x}, t)c(\mathbf{x}, t) - \sum_{m=1}^2 c_m(\mathbf{x}, t)q_m(t)w_m(\mathbf{x}), \quad (21)$$

where the first term on the right in Eq. (21) represents the rate of release of solute from a site in Ω , perhaps from the vessel wall; the second term represents the degradation of solute in the domain by chemical reactions; and the third term represents the addition or removal of solute by fluid entry or exit at the fluid sources within the model vessel. The rate of degradation, which is assumed to have first-order kinetics, may depend on the local structures, and therefore is characterized by a nonnegative coefficient $a(\mathbf{x}, t)$ that may vary in space and time. The concentration $c_m(\mathbf{x}, t)$ equals $c(\mathbf{x}, t)$ for exiting flow ($q_m(t) \leq 0$), however $c_m(\mathbf{x}, t)$ must be specified for entering flow ($q_m(t) > 0$). No solute is allowed to enter or leave the fluid domain by way of the external source/sink.

3. NUMERICAL METHODS

A computational grid is placed on Ω by defining grid points $\mathbf{x}_{ij} = (x_i, y_j) = (ih, jh)$, where $i = 0, 1, 2, \dots, 4J - 1$; $j = 0, 1, 2, \dots, J - 1$; and $h = L/J$ (in our case, L is $50 \mu\text{m}$). Time is discretized in steps of Δt . The analytical solution at each grid point (i.e., $\mathbf{u}(\mathbf{x}_{ij}, n\Delta t)$) is approximated by a numerical solution $\mathbf{u}_{ij}^n$; the same convention is applied to other variables—hence, e.g., p_{ij}^n , $\mathbf{F}_{ij}^n$, and c_{ij}^n .

The material points of the immersed boundary are discretized by conceptualizing the boundary as N Lagrangian boundary nodes, denoted by s_k for $k = 1, 2, \dots, N$. The nodes are configured, however, so that the first node is at the distal end of the model vessel (coinciding with $s = 0$), with even-numbered nodes increasing along the left side and odd-numbered nodes increasing along the right side. The location of boundary node s_k at time $n\Delta t$ is given by $\mathbf{X}_k^n$. In the unstressed configuration (at $t = 0$), adjoining nodes along the boundary (e.g., s_k and s_{k+2}) are separated by distance $\Delta s_{k\pm} = |\mathbf{X}_{k\pm 2}^0 - \mathbf{X}_k^0|$. Along the sides of the model arteriole $\Delta s_{k\pm} = h/2$; along the semicircular ends of the vessel, the nodal points are initially separated by the arc-length distance $\pi h/8$, as in Fig. 1. The length of the unstressed boundary segment associated with node s_k is $\Delta s_k = (\Delta s_{k+} + \Delta s_{k-})/2$.

An approximation $\delta_h(\mathbf{x})$ to the two-dimensional Dirac delta function $\delta(\mathbf{x})$, which is needed, e.g., in Eqs. (3) and (4), is given by Peskin and McQueen [29] as

$$\delta_h(\mathbf{x}) = h^{-2}\phi(x/h)\phi(y/h), \quad (22)$$

where the one-dimensional approximate Dirac delta function $\phi(r)$ is given by

$$\phi(r) = \begin{cases} \frac{1}{8}(3 - 2|r| + \sqrt{1 + 4|r| - 4r^2}), & |r| \leq 1, \\ \frac{1}{2} - \phi(2 - |r|), & 1 \leq |r| \leq 2, \\ 0, & |r| \geq 2. \end{cases} \quad (23)$$

This formulation of an approximate delta function has properties that are particularly appropriate for use in the IMBM [29].

The approximate delta function is also used to describe the source weighting functions. The discretization of $w_m(\mathbf{x})$ is given by $w_m(\mathbf{x}_{ij}) = \delta_{h_o}(\mathbf{x}_{ij} - \mathbf{x}_{sm})$, where $\mathbf{x}_{sm}$ is the center of the source or sink for $m = 1, 2$ and h_o is the coarsest grid spacing, which is $(50/32) \mu\text{m}$. (Note that the parameter h_o is fixed so that the physical sizes of the source, sink, and external source/sink are maintained when the space discretization is varied.) The weighting function over the external source/sink, $w_0(\mathbf{x})$, is discretized as the one-dimensional approximation to the Dirac delta function, normalized to have weight one; it is given by $\phi((y_i - y_{s0})/h_o)/200$.

At time $n\Delta t$, the node locations $\mathbf{X}_k^n$, fluid velocity $\mathbf{u}_{ij}^n$, and solute concentration c_{ij}^n are assumed known. The updated dependent variables ($\mathbf{u}_{ij}^{n+1}$, p_{ij}^{n+1} , c_{ij}^{n+1} , and $\mathbf{X}_k^{n+1}$) are computed by a first-order splitting in time, which may be summarized in four steps:

1. Compute the boundary force density $\mathbf{f}_k^n$ at each node s_k and spread this force density to the grid via the approximate delta function $\delta_h(\mathbf{x}_{ij})$ to obtain the corresponding grid force density $\mathbf{F}_{ij}^n$.
2. Using the grid force density $\mathbf{F}_{ij}^n$, solve the discretized Navier–Stokes equations to obtain the updated fluid velocity $\mathbf{u}_{ij}^{n+1}$ and pressure p_{ij}^{n+1} .
3. Use interpolation via the approximate delta function $\delta_h(\mathbf{x}_{ij})$ to determine the boundary node fluid velocity $\mathbf{U}_k^{n+1}$ from the nearby grid velocities $\mathbf{u}_{ij}^{n+1}$; move each node at the velocity $\mathbf{U}_k^{n+1}$ to obtain the updated boundary node configuration $\mathbf{X}_k^{n+1}$.
4. Compute the updated solute concentration c_{ij}^{n+1} by advection with the fluid velocity $\mathbf{u}_{ij}^{n+1}$; include solute diffusion and solute sources.

Each step will now be described in detail.

3.1. Force Determination

The boundary force component is given by a centered-difference discretization of Eqs. (10)–(12),

$$(\mathbf{f}_B)_k^n = \frac{K_B}{\Delta s_k} \left[\left(\left| \frac{\Delta \mathbf{X}_{k+}^n}{\Delta s_{k+}} \right| - 1 \right) (\boldsymbol{\tau}_B)_{k+}^n - \left(\left| \frac{\Delta \mathbf{X}_{k-}^n}{\Delta s_{k-}} \right| - 1 \right) (\boldsymbol{\tau}_B)_{k-}^n \right], \quad (24)$$

where $\Delta \mathbf{X}_{k+}^n = \mathbf{X}_{k+2}^n - \mathbf{X}_k^n$, $\Delta \mathbf{X}_{k-}^n = \mathbf{X}_k^n - \mathbf{X}_{k-2}^n$, and $(\boldsymbol{\tau}_B)_{k\pm}^n = \Delta \mathbf{X}_{k\pm}^n / |\Delta \mathbf{X}_{k\pm}^n|$.

For simplicity, each tether will be constructed to always act at a particular node. If a tether (fixed to a point $\mathbf{x}_T$) is attached to a node $\mathbf{X}_k^n$ (corresponding to the parametric location $s' = s_k$), then the tether force density (c.f. Eqs. (13)–(14)) is given by

$$(\mathbf{f}_T)_k^n = \frac{K_T}{\Delta s_k} \max\{0, (|\mathbf{x}_T - \mathbf{X}_k^n|) - (L_T)_k\} (\boldsymbol{\tau}_T)_k^n, \quad (25)$$

where the unit vector $(\boldsymbol{\tau}_T)_k^n$ for $\mathbf{x}_T \neq \mathbf{X}_k^n$ is given by

$$(\boldsymbol{\tau}_T)_k^n = \frac{\mathbf{x}_T - \mathbf{X}_k^n}{|\mathbf{x}_T - \mathbf{X}_k^n|}. \quad (26)$$

The myogenic component of the force density $(\mathbf{f}_M)_k^n$ on node s_k depends on the approximate pressure P_k^n and approximate concentration c_k^n (corresponding to $P(s_k, n\Delta t)$ and $c(\mathbf{X}(s_k, t), n\Delta t)$, respectively). The expression approximating the transmural pressure (Eq. (16)) is discretized as

$$P_k^n = h^2 \sum_{ij} p_{ij}^n (\delta_h(\mathbf{x}_{ij} - \mathbf{G}_k^n) - w_0(\mathbf{x}_{ij})). \quad (27)$$

The concentration at the boundary is computed by

$$c_k^n = h^2 \sum_{ij} c_{ij}^n \delta_h(\mathbf{x}_{ij} - \mathbf{X}_k^n). \quad (28)$$

Because of the symmetry of our labeling convention, the nodes labeled k and $k+1$, for even k , are located at the same x -location on opposite sides of the model arteriole; thus, $P_k^n = P_{k+1}^n$, $\mathbf{G}_k^n = \mathbf{G}_{k+1}^n = (\mathbf{X}_k^n + \mathbf{X}_{k+1}^n)/2$, and $(\mathbf{f}_M)_{k+1}^n = -(\mathbf{f}_M)_k^n$.

Once the force density component at each node has been computed, the total force density at time index n and node s_k is given by $\mathbf{f}_k^n = (\mathbf{f}_B)_k^n + (\mathbf{f}_T)_k^n + (\mathbf{f}_M)_k^n$. This force density is transferred to the nearby grid points by the discretized version of Eq. (3):

$$\mathbf{F}_{ij}^n = \sum_k \mathbf{f}_k^n \delta_h(\mathbf{x}_{ij} - \mathbf{X}_k^n) \Delta s_k. \quad (29)$$

3.2. Fluid Equations

The updated fluid velocity is computed in a tripartite splitting that is similar to that described in Ref. [29]. However, here we use a spectral method to evaluate the contributions of the viscous and incompressibility terms, i.e., we apply the continuous Fourier transform, evaluate unknown quantities algebraically in transform space, and then use the discrete Fourier transform (DFT) to return to physical space. In Ref. [29], the same terms are first discretized in physical space, and the DFT is then used to solve the discretized equations. The sources and sinks are handled much as in Ref. [27].

In the tripartite splitting, two provisional values, $\mathbf{u}_{ij}^{n+1,1}$ and $\mathbf{u}_{ij}^{n+1,2}$, are calculated before $\mathbf{u}_{ij}^{n+1}$ is finally determined. The first provisional value includes contributions from the force density $\mathbf{F}$, computed by an explicit forward Euler step, and from the damping term $-\sigma\kappa\mathbf{u}$, computed by an implicit backward Euler step, yielding

$$\mathbf{u}_{ij}^{n+1,1} = \frac{\mathbf{u}_{ij}^n + (\Delta t/\rho)\mathbf{F}_{ij}^n}{1 + (\Delta t/\rho)\kappa\sigma_{ij}^n}. \quad (30)$$

The values of σ_{ij}^n are obtained from σ , a linear superposition of hyperbolic tangent functions constructed to give a value of approximately one outside the vessel and zero inside. Along the gridlines x_i that intersect the boundary, σ_{ij}^n is given by

$$\begin{aligned} \sigma_{ij}^n &= \sigma(x_i, y_j, n\Delta t) \\ &= 1 - \frac{1}{2} [\tanh(0.8(y_j - (y_{ibl}^n - 2h_o))) + \tanh(0.8(y_j - (y_{ibr}^n + 2h_o)))], \end{aligned} \quad (31)$$

where y_{ibl}^n and y_{ibr}^n are the left and right y -coordinates of the intersection of $x = x_i$ with the immersed boundary, respectively, at time $n\Delta t$. For gridlines x_i that do not intersect the

immersed boundary, $\sigma = 1$. With this specification of σ , $\sigma_{ij}^n \approx 0.001$ at a distance $h_o/2$ inside the boundary, and σ_{ij}^n is orders of magnitude smaller for points further inside the boundary. At a distance $4h_o$ outside the boundary, σ_{ij}^n exceeds 0.99, and σ_{ij}^n rapidly approaches 1 as the distance from the boundary increases.

The second provisional value $\mathbf{u}_{ij}^{n+1,2} = (u_{ij}^{n+1,2}, v_{ij}^{n+1,2})$, which includes the contribution of the advection term $(\mathbf{u} \cdot \nabla)\mathbf{u}$, is obtained from $\mathbf{u}_{ij}^{n+1}$ by an upwind differencing,

$$\begin{aligned} u_{ij}^{n+1,2} = & u_{ij}^{n+1,1} - \frac{\Delta t}{h} [(u_+ (u_{ij}^{n+1,1} - u_{i-1,j}^{n+1,1}) + u_- (u_{i+1,j}^{n+1,1} - u_{ij}^{n+1,1})) \\ & + (v_+ (u_{ij}^{n+1,1} - u_{i,j-1}^{n+1,1}) + v_- (u_{i,j+1}^{n+1,1} - u_{ij}^{n+1,1}))], \end{aligned} \quad (32)$$

$$\begin{aligned} v_{ij}^{n+1,2} = & v_{ij}^{n+1,1} - \frac{\Delta t}{h} [(u_+ (v_{ij}^{n+1,1} - v_{i-1,j}^{n+1,1}) + u_- (v_{i+1,j}^{n+1,1} - v_{ij}^{n+1,1})) \\ & + (v_+ (v_{ij}^{n+1,1} - v_{i,j-1}^{n+1,1}) + v_- (v_{i,j+1}^{n+1,1} - v_{ij}^{n+1,1}))], \end{aligned} \quad (33)$$

where $u_{\pm} = \pm \max\{0, \pm u_{ij}^n\}$ and $v_{\pm} = \pm \max\{0, \pm v_{ij}^n\}$.

In the third and final step of the splitting, discretized forms of the equations

$$\rho \partial_t \mathbf{u} = -\nabla p + \mu \Delta \mathbf{u}, \quad (34)$$

$$\nabla \cdot \mathbf{u} = \sum_{m=1}^2 q_m(t)(w_m(\mathbf{x}) - w_0(\mathbf{x})) \quad (6)$$

are solved simultaneously for the updated fluid velocity $\mathbf{u}_{ij}^{n+1}$ and pressure p_{ij}^{n+1} . To incorporate the fluid sources, the fluid velocity and pressure are written as a superposition of a solution without fluid sources plus terms that include each source independently. For simplicity in the derivation of the appropriate equations, we will use equations that are continuous in space but discretized in time. Thus we will write the second provisional value $\mathbf{u}_{ij}^{n+1,2}$ as $\mathbf{u}^{n+1,2}(\mathbf{x})$, and we will write the time-advanced fluid velocity $\mathbf{u}_{ij}^{n+1}$ and pressure p_{ij}^{n+1} as $\mathbf{u}(\mathbf{x})$ and $p(\mathbf{x})$, respectively.

Now using superposition, we assume that

$$\mathbf{u}(\mathbf{x}) = \tilde{\mathbf{u}}(\mathbf{x}) + \sum_{m=1}^2 \mathbf{b}_m(\mathbf{x}) q_m^{n+1} \quad (35)$$

and

$$p(\mathbf{x}) = \tilde{p}(\mathbf{x}) + \sum_{m=1}^2 r_m(\mathbf{x}) q_m^{n+1}, \quad (36)$$

where the terms arising in the absence of sources, $\tilde{\mathbf{u}}(\mathbf{x})$ and $\tilde{p}(\mathbf{x})$, satisfy

$$\rho \left(\frac{\tilde{\mathbf{u}}(\mathbf{x}) - \mathbf{u}^{n+1,2}(\mathbf{x})}{\Delta t} \right) = -\nabla \tilde{p}(\mathbf{x}) + \mu \Delta \tilde{\mathbf{u}}(\mathbf{x}), \quad (37)$$

$$\nabla \cdot \tilde{\mathbf{u}}(\mathbf{x}) = 0, \quad (38)$$

and the terms on the far right in Eqs. (35) and (36) represent contributions from the sources.

The effects of the external source/sink ($m = 0$) are implicitly included in these equations through volume conservation in Ω .

If we substitute Eqs. (35) and (36) into Eq. (6) and into the time-discretized version of Eq. (34), and if we take into account, that, by assumption, $\tilde{\mathbf{u}}(\mathbf{x}, t)$ and $\tilde{p}(\mathbf{x}, t)$ satisfy Eqs. (37) and (38), we find that q_m^{n+1} , $\mathbf{b}_m(\mathbf{x})$, and $r_m(\mathbf{x})$ must satisfy

$$\sum_{m=1}^2 q_m^{n+1} \left[\rho \frac{\mathbf{b}_m(\mathbf{x})}{\Delta t} + \nabla r_m(\mathbf{x}) - \mu \Delta \mathbf{b}_m(\mathbf{x}) \right] = 0, \quad (39)$$

$$\sum_{m=1}^2 q_m^{n+1} [\nabla \cdot \mathbf{b}_m(\mathbf{x}) - (w_m(\mathbf{x}) - w_0(\mathbf{x}))] = 0. \quad (40)$$

Because the source magnitudes are independent of the flow vector $\mathbf{b}_m$, the terms within the square brackets in Eqs. (39) and (40) must vanish for each m . Therefore, $\mathbf{b}_m(\mathbf{x})$ and $r_m(\mathbf{x})$ must satisfy

$$\rho \left(\frac{\mathbf{b}_m(\mathbf{x}) - 0}{\Delta t} \right) = -\nabla r_m(\mathbf{x}) + \mu \Delta \mathbf{b}_m(\mathbf{x}), \quad (41)$$

$$\nabla \cdot \mathbf{b}_m(\mathbf{x}) = w_m(\mathbf{x}) - w_0(\mathbf{x}). \quad (42)$$

Thus the unknowns $\mathbf{b}_m(\mathbf{x})$, $r_m(\mathbf{x})$, $\tilde{\mathbf{u}}(\mathbf{x})$, and $\tilde{p}(\mathbf{x})$, in Eqs. (37), (38), (41), and (42), can all be found by solving a system of the form

$$\rho \left(\frac{\mathbf{u} - \mathbf{w}}{\Delta t} \right) = -\nabla p + \mu \Delta \mathbf{u}, \quad (43)$$

$$\nabla \cdot \mathbf{u} = S(\mathbf{x}), \quad (44)$$

where $\mathbf{w}$ and $S(\mathbf{x})$ are given. Moreover, since $\mathbf{b}_m(\mathbf{x})$ and $r_m(\mathbf{x})$ depend only on fixed source locations and the time step Δt , Eqs. (41) and (42) need be solved only once.

Equations (43) and (44) may be solved by means of the Fourier transform, which, for a scalar function g on the finite domain $[0, 4L] \times [0, L]$, is given by

$$\hat{g}(k_x, k_y) = \int_0^L \int_0^{4L} \exp\{-i(2\pi/L)[k_x(x/4) + k_y y]\} g(x, y) dx dy. \quad (45)$$

Using this definition, the transform of the scalar form of Eqs. (43) and (44) is given by

$$\rho \left(\frac{\hat{u} - \hat{w}}{\Delta t} \right) = -k_1 \hat{p} + \mu k^2 \hat{u}, \quad (46)$$

$$\rho \left(\frac{\hat{v} - \hat{z}}{\Delta t} \right) = -k_2 \hat{p} + \mu k^2 \hat{v}, \quad (47)$$

$$(k_1 \hat{u} + k_2 \hat{v}) = \hat{S}, \quad (48)$$

where $\mathbf{u} = (u, v)$, $\mathbf{w} = (w, z)$, $(k_1, k_2) = -i(2\pi/L)(k_x/4, k_y)$, and $k^2 = k_1^2 + k_2^2$. Solving this

system for $\hat{p}$, one obtains

$$\hat{p} = \frac{k_1 \hat{w} + k_2 \hat{z} - (1 - (\mu \Delta t / \rho) k^2) \hat{S}}{(\Delta t / \rho) k^2}. \quad (49)$$

Solving for $\hat{u}$ and $\hat{v}$ gives

$$\hat{u} = \frac{\hat{w} - (\Delta t / \rho) k_1 \hat{p}}{1 - (\mu \Delta t / \rho) k^2}, \quad (50)$$

$$\hat{v} = \frac{\hat{z} - (\Delta t / \rho) k_2 \hat{p}}{1 - (\mu \Delta t / \rho) k^2}. \quad (51)$$

Since $\hat{p}$ depends only on $\hat{w}$ and $\hat{S}$, and $\hat{u}$ depends only on $\hat{w}$ and $\hat{p}$, $\mathbf{u}$ and p can be determined by inverse transforming Eqs. (49), (50), and (51). Note that $\hat{p}$ is undefined when $k_1 = k_2 = 0$; we set $\hat{p}(0, 0) = 0$. This choice has no effect on the transformed velocities computed in Eqs. (50) and (51).

In our numerical realization of this solution method for Eqs. (43) and (44), we first compute, by means of the fast Fourier transform (FFT) [3, 30], the discrete Fourier transform (DFT) of $\mathbf{w}_{ij}$ and of S_{ij} . (Recall that the identities of $\mathbf{w}$ and S depend on which set of unknowns is being solved for: $(\mathbf{b}_1, r_1)$, $(\mathbf{b}_2, r_2)$, or $(\hat{\mathbf{u}}, \hat{p})$; see Eqs. (37), (38), (41), and (42).) Then a discretized $\hat{p}$ is determined from Eq. (49), and discretized $\hat{u}$ and $\hat{v}$ are determined from Eqs. (50) and (51), respectively. Finally, the FFT is used to compute the inverse DFTs of these variables, i.e., to compute p_{ij} and $\mathbf{u}_{ij}$.

We now show how to compute the source flow rates q_1^{n+1} and q_2^{n+1} for use in Eqs. (35) and (36). If Eq. (36) is substituted into Eq. (9), and the resulting expression for P_m is substituted into Eq. (8), and if the resulting expression is written in discretized form, one obtains

$$\begin{aligned} q_m^{n+1} R_m + h^2 \sum_{k=1}^2 \sum_{ij} r_{k,ij} q_k^{n+1} (w_m(\mathbf{x}_{ij}) - w_0(\mathbf{x}_{ij})) \\ = \bar{P}_m - h^2 \sum_{ij} \tilde{p}_{ij}^{n+1} (w_m(\mathbf{x}_{ij}) - w_0(\mathbf{x}_{ij})), \end{aligned} \quad (52)$$

which is equivalent to a linear system of the form

$$A Q^{n+1} = B^{n+1}, \quad (53)$$

where

$$\begin{aligned} Q_m^{n+1} &= q_m^{n+1}, & m &= 1, 2, \\ B_m^{n+1} &= \bar{P}_m - h^2 \sum_{ij} \tilde{p}_{ij}^{n+1} (w_m(\mathbf{x}_{ij}) - w_0(\mathbf{x}_{ij})), & m &= 1, 2, \\ A_{mk} &= R_m \delta_{mk} + h^2 \sum_{ij} r_{k,ij} (w_m(\mathbf{x}_{ij}) - w_0(\mathbf{x}_{ij})), & m &= 1, 2; \quad k = 1, 2, \end{aligned} \quad (54)$$

and δ_{mk} is the Kronecker delta. Since $\tilde{p}_{ij}^n$ and $r_{m,ij}$ have already been computed, this system may be readily solved for Q^{n+1} .

Now, since $\tilde{\mathbf{u}}_{ij}^n$, $\tilde{p}_{ij}^{n+1}$, $\mathbf{b}_{m,ij}$, $r_{m,ij}$, and q_m^{n+1} are known, the updated velocity $\mathbf{u}_{ij}^{n+1}$ and pressure p_{ij}^{n+1} may be computed from the discretized versions of Eqs. (35) and (36), given by

$$\mathbf{u}_{ij}^{n+1} = \tilde{\mathbf{u}}_{ij}^n + \sum_{m=1}^2 \mathbf{b}_{m,ij} q_m^{n+1} \quad (55)$$

and

$$p_{ij}^{n+1} = \tilde{p}_{ij}^{n+1} + \sum_{m=1}^2 r_{m,ij} q_m^{n+1}. \quad (56)$$

So that the pressure displayed in the figures would be compatible with experimental usage, the pressure computed through Eq. (56) was translated so that the average exterior pressure is near zero and the interior pressure approximates the transmural pressure. The translation was effected by averaging the pressure over the generalized source/sink and subtracting that average from the pressure computed in Eq. (56). The resulting pressure exterior to the vessel falls between -1 and 1 mm Hg. The translated pressures are used in the figures.

3.3. Boundary Nodes

The local fluid velocity at node $\mathbf{X}_k^n$ is computed by interpolation from the velocity at nearby grid points; thus

$$\mathbf{U}_k^{n+1} = h^2 \sum_{ij} \mathbf{u}_{ij}^n \delta_h(\mathbf{x}_{ij} - \mathbf{X}_k^n). \quad (57)$$

The boundary node location is then updated by a forward difference, given by

$$\mathbf{X}_k^{n+1} = \mathbf{X}_k^n + \Delta t \mathbf{U}_k^{n+1}, \quad (58)$$

the discretized version of Eq. (4).

3.4. Solute

The solute equation, Eq. (20), is solved by a tripartite splitting. The advection step, $c_t + \nabla \cdot (\mathbf{u}c) = 0$, is solved using a multidimensional flux-corrected transport (FCT) method, with an eighth-order Richtmyer two-step Lax–Wendroff scheme as the high-order scheme and upwind differencing as the low-order scheme [37]. Zalesak has shown that this method provides good shape preservation when applied to the linear advection problem [38].

The diffusion step, $c_t = \nabla \cdot (D(\mathbf{x})\nabla c)$, may be solved using backward-time, centered-in-space differencing, and an iterative scheme, e.g., Richardson iteration [20]. However, in the computational examples presented below, D is assumed to be constant, so the system is easily solved by means of the Fourier transform. Thus, if $c^{n+1,1}$ is the concentration computed by the advection step, the second provisional concentration, $c^{n+1,2}$, may be computed by the backward difference

$$\frac{c^{n+1,2}(\mathbf{x}) - c^{n+1,1}(\mathbf{x})}{\Delta t} = D \Delta c^{n+1,2}(\mathbf{x}). \quad (59)$$

This equation may be Fourier transformed in space (using Eq. (45)), solved for $\widehat{c^{n+1,2}}$, and inverse transformed to obtain $c^{n+1,2}$. In practice, the discrete transforms are computed by means of the FFT.

Finally, the solute source step, expressed as $c_t = Q_S$, is solved by the forward Euler method to obtain c_{ij}^{n+1} .

4. AFFERENT ARTERIOLE

Thus far, the model has been described in general terms for application to a generic arteriole. However, in numerical simulations, we will represent an afferent arteriole in the kidney of a rat. A typical rat afferent arteriole is approximately $20\ \mu\text{m}$ in diameter and is $100\text{--}300\ \mu\text{m}$ in length. Flow in the afferent arteriole is regulated in large measure by the myogenic response of the walls and by vasoactive substances that modulate the contractility of vascular smooth muscle. The model afferent arteriole developed here includes the myogenic response and the action of the vasodilator nitric oxide (NO), which is released by the endothelial and smooth muscle cells that make up the vessel wall and which undergoes advection, diffusion, and degradation in the fluid flow.

Two aspects of the myogenic response will be included in the myogenic force density component $\mathbf{f}_M(s, t)$: a steady-state response that reduces vessel diameter as steady-state perfusion pressure increases (see Fig. 3), and a transient response to an increase in perfusion pressure that initially allows the vessel diameter to increase elastically, but subsequently initiates a muscular contraction, leading to a decrease in vessel diameter (see Fig. 4).

The myogenic response was computed from a phenomenological model in which the magnitude of the myogenic force component is given by

$$M(s, t) = K_M(s, t) \frac{D(s, t)}{D_0} + \gamma \frac{\partial_t D(s, t)}{D_0}, \quad (60)$$

where $D(s, t) = |\mathbf{X}(s, t) - \mathbf{X}(-s, t)|$ is vessel diameter, $K_M(s, t)$ is a myogenic force coefficient, γ is a constant coefficient of viscosity, and D_0 is a reference diameter. Equation (60) is based on the Voigt model for a viscoelastic material; the first term is suggested by Laplace's law and the second term represents the viscous response of the tissue [16].

The coefficient $K_M(s, t)$ is constructed as a linear combination of a coefficient $K_a(s, t)$ for fully active myogenic response and a coefficient $K_p(s, t)$ for a fully passive (i.e., elastic) vessel,

$$K_M = (K_a - K_p)(1 - r) + K_p, \quad (61)$$

where r depends on the concentration of NO along the vessel wall, $c(\mathbf{X}(s, t), t)$. The fully active myogenic coefficient is the solution of a second-order differential equation that permits a transient response,

$$b_3 \partial_t^2 K_a + b_2 \partial_t K_a + b_1(s, t) K_a = P(s, t), \quad (62)$$

where $P(s, t)$ is the transmural pressure.

The coefficient b_1 , which models the steady-state response, is given by

$$b_1(P(s, t)) = \frac{22 - d(s)(P(s, t) - 50)}{D_0}, \quad (63)$$

where $d(s)$ increases along the active portion of the model arteriole. Specifically, as s varies from the attachment point of tether T6 to the attachment point of T8, $d(s)$ increases linearly from a value of 0.1 to a value of 0.3. Because intravascular pressure decreases along the flow direction, the scaling provided by $d(s)$ is necessary for the model arteriole to maintain a nearly constant diameter along its length, as do afferent arterioles in vivo. From this result one may reasonably hypothesize that the myogenic response in vivo is spatially inhomogeneous.

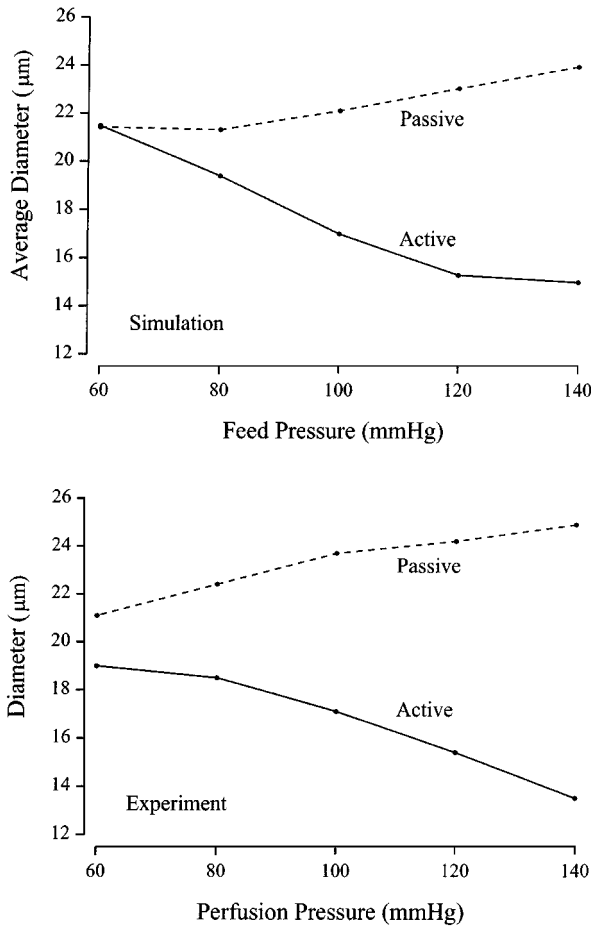


FIG. 3. Comparison of steady-state myogenic responses based on our simulations (upper panel) with experimental responses from a rat afferent arteriole (lower panel). In each panel, arteriolar diameter is given as a function of pressure (simulation feed pressure $\bar{P}_1$ slightly exceeds simulation perfusion pressure). Also, in each panel, the dashed line represents the passive arteriolar dilation resulting when the myogenic response has been disabled, and the solid line represents myogenic contraction. The upper panel shows the average diameter between $x = 50$ and $150 \mu\text{m}$ for quasi-steady-state flows with spatial grid refinement $h_o/2$ (see Section 5). The lower panel shows the response of rat afferent arterioles at mid-length, as feed pressure increases from 60 to 140 mm Hg. (Experimental data is from published work of Casellas and Moore [6]).

The coefficients γ , b_2 , and b_3 , which are required for a description of the dynamic response, were obtained as follows. Equations (60) and (62) were considered to be independent of s , $d(s)$ in Eq. (63) was set to 0.1, and a pressure $P(t) = 60 + 40(1 - \exp^{-1.12t})$ mm Hg was prescribed. This pressure was converted into units of force to obtain $M(t)$ in Eq. (60), and the equations were solved for the resulting diameter $D(t)$, using trial values of the unknown coefficients. Values for the coefficients were determined, by means of a least-squares algorithm, so that the time course of the diameter closely approximated previously unpublished experimental data from the laboratory of Leon Moore. The coefficients so obtained are given in Table III. Figure 4 shows the experimental values for arteriolar diameter, the solution obtained from the model when Eqs. (60) and (62) were solved as just described using the coefficients in Table III, and the diameter obtained for the full simulation, incorporating Eqs. (60), (62), and (63). In the full simulation, $P(s, t)$ was computed through the fluid

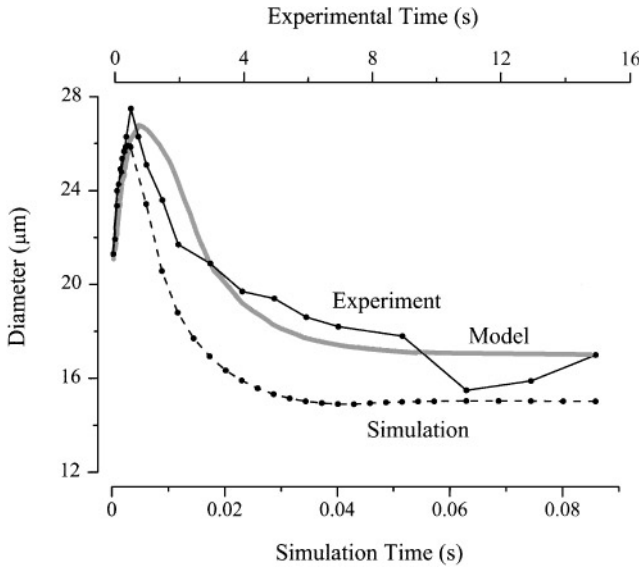


FIG. 4. Comparison of transient myogenic response for a rat afferent arteriole, the phenomenological model of the myogenic response, and the full simulation. Solid line denotes rat afferent arteriolar response when perfusion pressure was increased from 60 to 100 mm Hg. Experimental time (in seconds) is shown across the top of the diagram. The wide gray line shows the model diameter computed through Eqs. (60) and (62) to determine the dynamic coefficients (see text). The dashed line shows the response of the simulation at $x \approx 50 \mu\text{m}$ for the spatial grid refinement $h_o/2$, after the feed pressure was increased stepwise from 80 to 120 mm Hg. The time axis for the simulation is shown at the bottom of the figure; the time scale for the simulation was increased by a factor of 175 (by scaling the speed of the myogenic response) so that physiological times could be attained in the numerical simulations. The data labeled “Experiment” is previously unpublished data obtained in the laboratory of Leon Moore.

equations, and $M(s, t)$ depended on the diameter that resulted from the movement of the immersed boundary in the fluid. The curve for the simulation was computed as the distance between nodes located near $x = 50 \mu\text{m}$, where the steady-state pressure was about 100 mm Hg. Because the simulation is spatially inhomogeneous, a more complex problem was being solved than in the simple model used to estimate the parameters for Eq. (62); consequently the simulation diameter is somewhat less than the model over most of the time course exhibited in Fig. 4.

The time scale for the simulation of the dynamic myogenic response was increased by a factor of 175 over the time scale of the experimental response. This change of scale was introduced because the physiological response requires more time than can be practically simulated using the IMBM as formulated here; however, the change of scale can be justified by a consideration of characteristic times (see DISCUSSION). The coefficients γ , b_2 , and b_3 determined above incorporate the change of scale.

To simulate observed experimental diameters of a passive rat afferent arteriole (see Fig. 3), the coefficient $K_p(P(s, t))$ for a passive model arteriole was determined from the steady-state portion of Eq. (60). The force M was identified with the pressure P , and K_M was taken to be K_p . An experimental relationship between pressure and diameter, taken from Ref. [6] and reproduced in Fig. 3 (Experiment: Passive), was found to be well-approximated by

$$K_p = 50 + 2(P(s, t) - 50)^{0.8}, \quad (64)$$

where $P(s, t)$ is in millimeters of mercury. Figure 3 shows the passive response, for the full simulation, with this specification of K_p .

The percentage of relaxation at a particular vasodilator concentration, which can be determined from a dose–response curve, gives the fractional relaxation, r , as a function of local solute concentration. For NO, this relationship was approximated by

$$r(c(X(s, t), t)) = \frac{1}{2}[1 + \tanh(\pi[\log_{10}(c(X(s, t), t)/0.0004) - 1])] \tag{65}$$

which gives practically no relaxation at a concentration of $0.0004 \mu\text{M}$ ($r \approx 0.00186$) and nearly complete relaxation at a concentration of $0.04 \mu\text{M}$ ($r \approx 0.998$). This relationship was based on data from Ref. [21].

In the numerical implementation of the myogenic force component, Eq. (62) is recast as two first-order differential equations and solved at each node s_k by the forward Euler method for $(K_a)_k^n$. The NO concentration at each node c_k^n is computed via Eq. (28), and the average of c_k^n and c_{k+1}^n is used in Eq. (65) to determine the fractional relaxation.

The myogenic component of the force density at node s_k is then calculated as

$$(\mathbf{f}_M)_k^n = \left((K_M)_k^n \frac{D_k^n}{D_0} + \gamma \frac{(D_k^n - D_k^{n-1})}{D_0 \Delta t} \right) \frac{\mathbf{X}_{k+1}^n - \mathbf{X}_k^n}{|\mathbf{X}_{k+1}^n - \mathbf{X}_k^n|} \tag{66}$$

for k even. Because nodes s_k and s_{k+1} are opposing, $(\mathbf{f}_M)_{k+1}^n = -(\mathbf{f}_M)_k^n$ for k even.

5. NUMERICAL SIMULATIONS

Parameter selection. Parameters must be specified to characterize the fluid (i.e., the blood), the configuration and physical properties of the vessel, and the solute NO. Parameter selection criteria are summarized below; parameters are collected in Table III.

TABLE III
Model Parameters

Parameter	Symbol	Value	Reference
Fluid density	ρ	1.055 gm/cm ²	[1]
Apparent dynamic blood viscosity	μ	1.75×10^{-2} gm/(cm · s)	[23]
Damping coefficient	κ	875,000 gm/(cm ² · s)	
Lateral spring coefficient	K_B	3,104,307 dyne/cm	
Tether spring coefficient	K_T	10,000 dyne/cm	
External pressure at fluid source	$\bar{P}_1$	100 mm Hg	
External pressure at fluid sink	$\bar{P}_2$	0 mm Hg	
Resistance at fluid source	R_1	2.0×10^7 gm/(s · cm ²)	
Resistance at fluid sink	R_2	2.0×10^8 gm/(s · cm ²)	
Myogenic model viscosity coefficient	γ	0.193 mm Hg · s	
Myogenic model coefficient	b_2	1.60 s	
Myogenic model coefficient	b_3	0.282 s ²	
NO release rate	s_0	8.32 nmol/(cm ³ · s)	^a
NO diffusivity	D	3.3×10^{-5} cm ² /s	[25]
NO removal coefficient	a_1	3.96×10^{-3} s ⁻¹	[26]
NO–hemoglobin binding coefficient	a_2	0.198 s ⁻¹	[22]

^a Based on the study reported in Ref. [36] (Mark Vaughn, personal communication).

Blood is a complex non-Newtonian fluid composed of plasma, red blood cells, white blood cells, and platelets. To obtain a tractable model, however, blood was simulated as a Newtonian fluid with an apparent dynamic viscosity. Experimental measurements in an arteriole of diameter $23\text{ }\mu\text{m}$ in cat mesentery indicate an apparent viscosity of $1.75 \times 10^{-2}\text{ gm}/(\text{cm} \cdot \text{s})$ [23]. (The viscosity of water at body temperature is $\approx 0.725 \times 10^{-2}\text{ gm}/(\text{cm} \cdot \text{s})$).

The damping coefficient κ was chosen empirically to damp fluid motion exterior to the model vessel. The spring coefficients K_B and K_T were chosen empirically to contain the pressures in the physiological range and to anchor the model vessel over the vessel source and sink.

The external pressure at the fluid source, $\bar{P}_1 = 100\text{ mm Hg}$, and the external resistance at the fluid source, $R_1 = 2 \times 10^7\text{ gm}/(\text{s} \cdot \text{cm}^2)$, were chosen so that the pressure at the fluid source would be approximately 95 mm Hg . The external pressure at the fluid sink, $\bar{P}_2 = 0\text{ mm Hg}$, and the external resistance at the fluid sink, $R_2 = 2 \times 10^8\text{ gm}/(\text{s} \cdot \text{cm}^2)$, were chosen so that the pressure at the fluid sink would be approximately 50 mm Hg . These values are consistent with pressures found in mid-juxtamedullary afferent arterioles in rat [6]. The myogenic model coefficients γ , b_2 , and b_3 were determined as described in Section 4.

The NO release rate, per unit area, from the wall of rabbit aorta, has been estimated [36; Mark Vaughn, personal communication] from experimental data [25] to be $5.2 \times 10^{-14}\text{ }\mu\text{mol}/(\mu\text{m}^2 \cdot \text{s})$. In the simulation, NO was released from an approximate delta function with side $4h_o$ centered on the vessel wall. For a fluid domain of any given thickness H , the rate of release of NO from the wall segment covered by the delta function (s_0 in Eq. (21)), equals the area of the vessel wall of height H enclosed by the delta function, times the release rate per unit area, divided by the volume enclosed by the box of height H ; that is,

$$s_0 = \frac{4h_o H \times \text{release rate}/(\text{unit area})}{(4h_o)^2 H} = 8.32\text{ nmol}/(\text{cm}^3 \cdot \text{s}). \quad (67)$$

NO diffusivity has been measured to be $3.3 \times 10^{-5}\text{ cm}^2/\text{s}$ at 37°C in rabbit aorta, and evidence indicates that NO diffuses freely in aqueous solutions and across biological membranes [25]. The rate of NO degradation, through reaction with oxygen and water to produce nitrites and nitrates, was based on a half-life of 3 s [26]. Based on a calculation that indicated that the rate of NO reaction with hemoglobin is more than 100 times more rapid than the rate of reaction with O_2 [22], the rate of NO-hemoglobin binding was assumed to be 50 times faster than the reaction of NO with water and dissolved oxygen.

Discretization in space and time. Simulations were conducted with $J = 32, 64$, and 128 points along the y -axis; thus $h = 50/32, 50/64$, or $50/128\text{ }\mu\text{m}$. As already specified in Section 3, $h_o = 50/32$, and the two grid refinements will be referred to as $h_o/2$ and $h_o/4$. Empirical studies indicate that a time step of $\Delta t = (5/175) \times 10^{-5}\text{ s}$ or smaller is necessary to maintain numerical stability, and except where stated otherwise, this time step was used. Simulations were conducted in double precision on Sun[®] Workstations or in single precision, using one processor on the Cray[®] T-90 supercomputer at the North Carolina Supercomputing Center. (Single precision on the Cray[®] is essentially equivalent to double precision on the Sun[®] Workstation.) A single numerical iteration that includes the full functionality of the simulation required approximately 0.025 CPU s on the Cray[®] supercomputer.

Machine error. Although the model prescribes a line of symmetry running through the center of the model arteriole ($y = 31.25\text{ }\mu\text{m}$) and although the myogenic forces acting on

corresponding nodes (s_k and s_{k-1}) are equal and opposite, we observed a loss of symmetry in boundary node locations. Investigation showed that the loss of symmetry arises from reductions in the number of significant digits in the subtractions that compute the boundary force component (i.e., in the evaluation of Eq. (24)), coupled with asymmetric machine roundoff. The loss of symmetry in boundary node locations results in a loss of symmetry in the force density applied to the vessel wall by the myogenic components. Eventually a region of reduced force density along one side of the model vessel results in an outward curvature of the wall; this curvature is accompanied by an inward curvature on the other side of the model vessel. When the curvature becomes sufficiently large, increased nodal separation results in a loss of fluid containment.

The accumulation of asymmetry was controlled by adjusting the computation of the force density pairs $(f_M)_{k+1}^n, (f_M)_k^n$: in response to any deviation from symmetry, a weighting of force components was applied to these force densities in the direction needed to restore symmetry. The weighting was constructed to correct both translational and rotational asymmetries.

The adjustment in weighting for the myogenic force density pairs was determined in the following way. Assume that node s_k has Eulerian position $\mathbf{X}_k = (X_k, Y_k)$ and node s_{k+1} has position $\mathbf{X}_{k+1} = (X_{k+1}, Y_{k+1})$. The line segment connecting these nodes crosses the line of symmetry at the point, say (p_x, p_y) , where $p_y = 31.25 \mu\text{m}$. Let $(f_M)_{k,y}^n$ be the y -component of the myogenic force density at node s_k , computed via Eq. (66). Then the adjusted component was computed as

$$\frac{2|Y_k - p_y|}{|Y_{k+1} - Y_k|} (f_M)_{k,y}^n, \quad (68)$$

and the adjustment for $(f_M)_{k+1,y}^n$ was computed analogously. Thus the node more distant from the centerline received more force than the other node, resulting in a correction toward the line of symmetry.

Similarly, if $(f_M)_{k,x}^n$ was the computed x -component of the myogenic force density at node s_k , the adjusted component was computed as

$$\frac{2|X_k - p_x|}{|X_{k+1} - X_k|} (f_M)_{k,x}^n, \quad (69)$$

and the adjustment for $(f_M)_{k+1,x}^n$ was computed analogously. Since the signs of $(f_M)_{k,x}^n$ and $(f_M)_{k+1,x}^n$ differ if there is any rotation relative to the y -axis, the adjusted x -components applied a corrective torque. The magnitude of these corrections was small relative to the myogenic force density; at quasi-steady-state (see below) the ratio of the myogenic force density to the adjustment force was more than 10^{10} .

Overview of simulation results. The following subsections describe an extensive series of tests that were performed to evaluate the properties, accuracy, and convergence of the numerical methods. For quasi-steady-state (see below), the flow profile in the model vessel has the expected parabolic shape for planar Poiseuille flow; the effective width of the model vessel wall is about $2h$. Shear stress magnitude at the model wall is consistent with values expected for planar Poiseuille flow. For a sufficiently refined spatial grid (e.g., $h_o/4$), the method exhibits robust first-order temporal convergence as the time step is refined. However, while vessel pressure and diameter profiles support convergence in space as the spatial grid is refined, the attainment of good convergence throughout the fluid domain

appears to be limited by the relatively coarse grids that are currently practicable. For good spatial resolution in the representation of the flow velocity profile, shear stress, and solute advection and diffusion, a grid refinement with $h < 1 \mu\text{m}$ is needed.

5.1. The Pressurized Vessel

The first test of the method was to compute a quasi-steady-state configuration for the wall and fluid flow and to compare the velocity in the x -direction with the parabolic velocity profile for planar Poiseuille flow. We say quasi-steady-state to indicate that the input and output flows are very nearly constant and equal; a pure steady state is not attainable because the walls continue to undergo small motions for long times. Beginning from a completely quiescent state, approximately 0.06 s are required to reach a quasi-steady-state.

Fluid flow rates at the source and sink, and the sum of these flow rates as a function of time for the three levels of spatial grid refinement are shown in Fig. 5. Flow rates associated with the grid refinement spacing $h = h_o$ undergo oscillations between ~ 0.02 and ~ 0.03 s; oscillations are of larger magnitude at the sink. These oscillations are associated with vessel diameter oscillations in the proximity of the sink. Oscillations are not present in the more spatially refined simulations.

The empirical order of convergence of source and sink flows at 0.07 s, based on values found for the three different spatial grid refinements, h_o , $h_o/2$, and $h_o/4$, was found to be ~ 1.0 . The order of convergence, here and in subsequent cases, was computed as follows. Let V represent the value of a dependent variable at a specific grid point, and suppose that

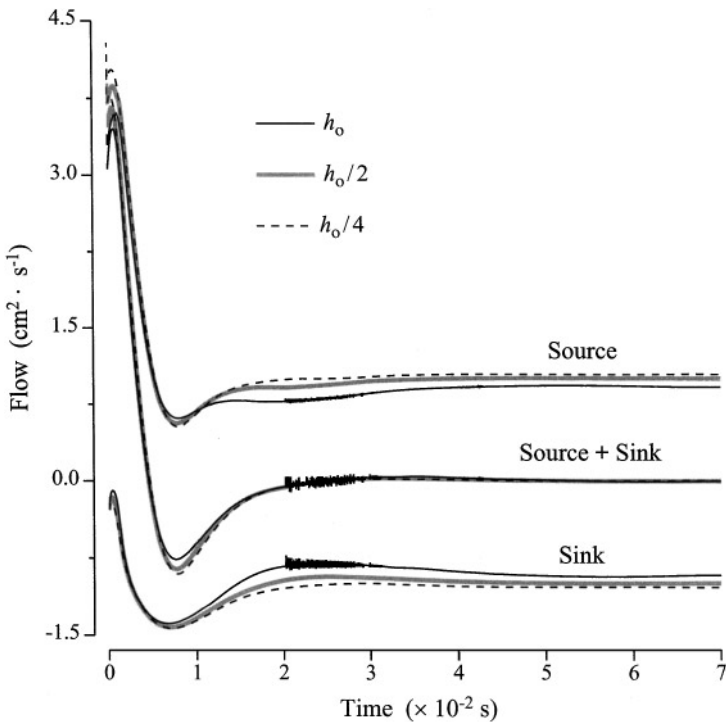


FIG. 5. Source and sink fluid flow rates and the sum of these rates are shown as a function of time for the three levels of spatial grid refinement. Flow for the least refined case exhibits oscillations (notably from ~ 0.02 to ~ 0.03 s) that are associated with oscillations in the boundary configuration near the sink.

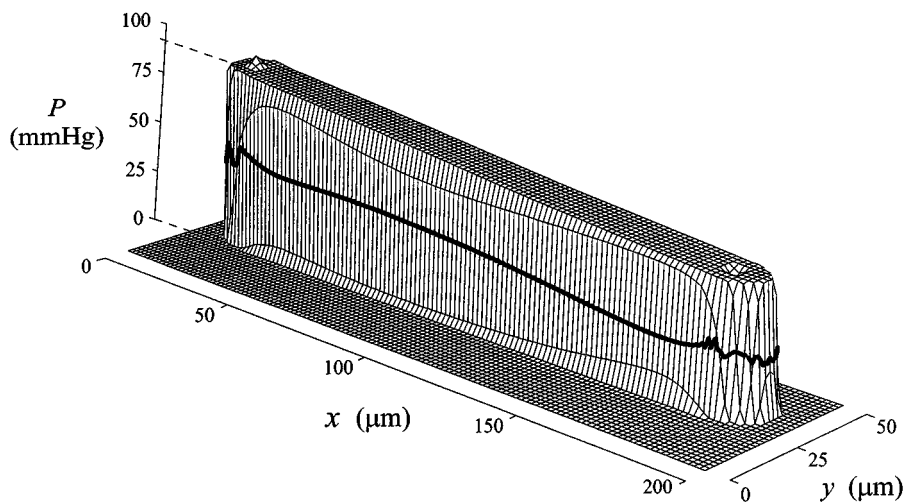


FIG. 6. Pressure as a function of position on the domain Ω for the quasi-steady-state is shown for the case $h = h_o/2$. Intersections of the wire-frame diagram correspond to every other value on the computational grid. The heavy solid line indicates the pressure at the center of the boundary that forms the model arteriolar wall. The sites of the source and sink are marked by local pressure deviations at the high and low pressure ends, respectively.

numerical approximations of V are given by values V_i , corresponding to space-step sizes $h_o/2^i$. An empirical measure of convergence of three strictly monotonic values V_i , $i = 0, 1, 2$, may be obtained by solving the three simultaneous equations

$$V_i = V_\infty + K (h_o/2^i)^p, \quad i = 0, 1, 2 \tag{70}$$

for p , V_∞ , and K (see Appendix). The empirical order of convergence is given by p , V_∞ is the value to which the V_i 's are (apparently) converging, and K is a measure of truncation error.

Figure 6 shows pressure (in millimeters of mercury) as a function of position on the domain Ω for the quasi-steady-state configuration with intermediate spatial grid refinement, $h = h_o/2$. The intersections of the wire-frame diagram correspond to every other value on the computational grid. The heavy solid line indicates the pressure along the centerline of the immersed boundary that forms the model arteriolar wall. There are small but discernable pressure variations over the fluid source and sink, but in general, the pressure is nearly constant across the width of the vessel and decreases linearly along the x -axis. At the vessel wall, however, pressure rapidly decreases to the external pressure, which is essentially zero.

The computed fluid velocity along the horizontal cross section $x = 100 \mu\text{m}$ was compared to the analogous analytical fluid velocity profiles for planar Poiseuille flow between two stationary parallel plates separated by width $2R$. The analytical velocity profile has the form $u_a(r) = (P_x/2\mu)(R^2 - r^2)$, where r is the location between the plates relative to the centerline (thus, $-R \leq r \leq R$), P_x is the pressure gradient, μ is the dynamic viscosity, and u_a is the velocity parallel to the plates [7].

For our comparisons, we approximated P_x from the simulation's pressure gradient by using a centered difference across $x = 100 \mu\text{m}$. The analytic velocity profile was evaluated for three different separation widths: (1) $2R$ equals the diameter D of the model vessel, from node center to node center; (2) $2R = D - 2h$; and (3) $2R = D - 4h$. Separate comparisons were made for the three levels of spatial grid refinement.

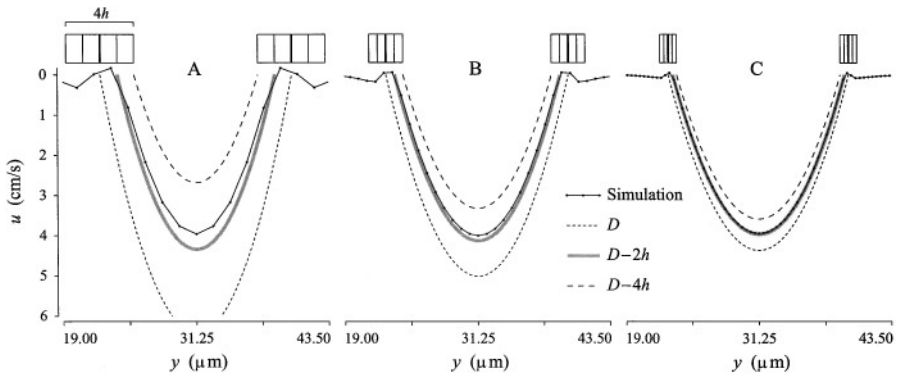


FIG. 7. Comparison of model fluid x -velocity u with analytical velocity profiles u_a for planar Poiseuille flow, along the cross section at $x = 100 \mu\text{m}$ for the three levels of spatial refinement (h_o , $h_o/2$, $h_o/4$, left to right). The solid lines indicate model velocity u ; the dots indicate grid points. Boxes above the profiles show the locations of boundary nodes (heavy lines) and the wall thicknesses, based on the support of the approximate delta function (width $4h$). Analytical profiles are shown for three different diameters: (1) D , the diameter of the model arteriole based on the node locations (short dashes); (2) $D - 2h$ (wide gray line); and (3) $D - 4h$ (long dashes). This figure indicates that the effective diameter of the model vessel is $D - 2h$. This figure also shows that the velocity profiles exhibit oscillations of decreasing magnitude in the vicinity of the boundary. The oscillations appear to be attributable, at least in part, to a Gibbs-type phenomenon that arises because in the analytic limit the derivative of the velocity would not be continuous at the boundary.

Figure 7, panels A, B, and C, show the comparisons for the three levels of spatial grid refinement. Boxes above the profiles show locations of the boundary nodes as heavy lines and they also show the wall thickness, based on a width of $4h$, which is the support of the approximate delta function. These results indicate that the model arteriolar wall has an effective thickness that depends on the support of the approximate delta function. The best fit between the model and analytical data was found when the separation width $2R$ equals $D - 2h$, which shows that the effective wall thickness is nearly $2h$. Indeed, the computed solution appears to converge to the corresponding analytical solution as the grid is refined.

Figure 7 also shows an oscillation in computed x -velocity near each boundary; the magnitude of the oscillation decreases with increasing spatial refinement. By comparing the simulated velocity with partial sums of the Fourier series approximation to the analytical velocity, we concluded that the oscillation is attributable, at least in part, to a Gibbs-type phenomenon that arises because the simulations are approximating a velocity profile that has a discontinuous derivative at the boundary.

In Fig. 8, the magnitude of computed shear stress is compared to the analytical shear stress for planar Poiseuille flow. Computed shear stress magnitude at $x = 100 \mu\text{m}$ is given as a function of cross-sectional distance for three levels of fluid refinement. Shear stress was calculated as $\mu(\partial u / \partial y)$, where $\mathbf{u} = (u, v)$; the derivative was computed with a centered-difference approximation. Analytical shear stress is given by $a P_x (y/a - 1/2)$, where a is the diameter, y is the cross-sectional distance, and P_x is the pressure gradient. For our comparison, P_x was determined using a centered-difference approximation along the model vessel's centerline. For each level of refinement, the diameter a was based on the diameter between the nodes of the boundary; however, other diameters based on the diameter between nodes minus $2h$ or $4h$ do not substantially change the analytical shear stress magnitude as a function of distance from the centerline. Vertical shaded bars show the support of the approximate delta function that forms the boundary. In all cases, the support is $4h$ wide

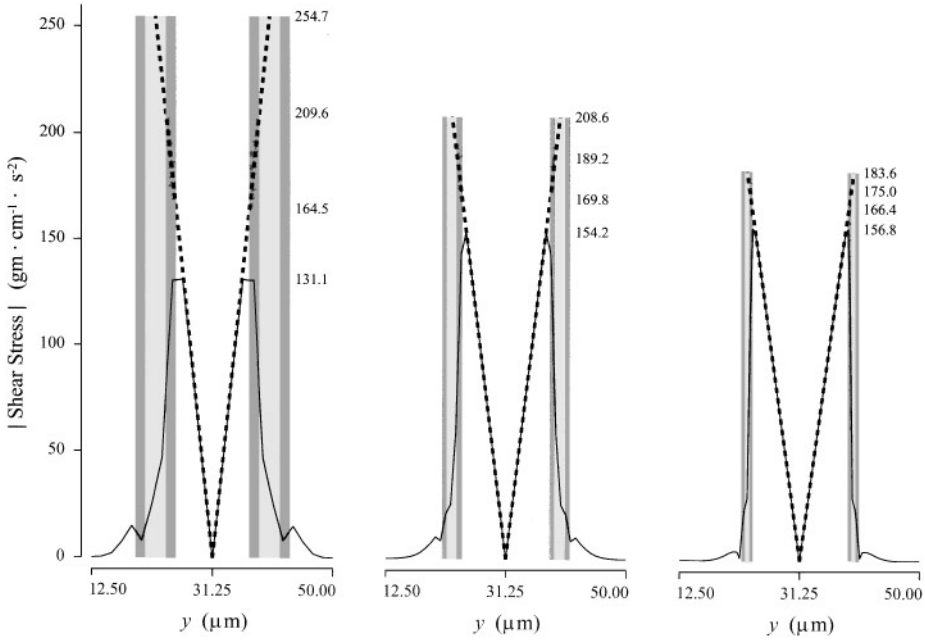


FIG. 8. Solid lines indicate the magnitude of shear stress as a function of the cross-sectional location at $x = 100 \mu\text{m}$ for the three spatial grid refinements (h_o , $h_o/2$, $h_o/4$, left to right). Dashed lines show the shear stress for planar Poiseuille flow, based on the local pressure gradient. Each set of vertical shaded bars shows the support of the approximate delta function that forms a boundary. Each boundary has support thickness $4h$, with heavier weight in the lightly shaded central region of thickness $2h$. The first three values along the right of each panel, from top to bottom, indicate the analytical shear stress magnitudes at the wall center (based on node locations), at the interior edge of the wall with thickness $2h$, and at the interior edge of the wall with thickness $4h$. The bottom value is the maximum shear stress magnitude computed by the simulation along the cross section.

and has heavier weight in the central $2h$ region (shaded lighter). Values along the right of each panel (from top to bottom) indicate the analytical shear stress magnitude at the wall center (based on node location), at the interior edge of the wall with width $2h$, and at the interior edge of the wall with width $4h$. The bottom value is the maximum shear stress magnitude computed by the simulation. As the spatial resolution is increased, maximum simulated shear stress magnitude along the cross section is converging to the analytical shear stress magnitude at the wall center with order 1.326. Rates of convergence to the other wall locations are similar. In future simulation studies, knowledge of this convergence rate will be essential for modeling NO release at the arteriolar wall, because NO release rate is a function of shear stress magnitude [4] (see DISCUSSION).

The larger shear stress magnitude near the wall computed for spatial grid refinement $h = h_o/4$ tends to stretch the immersed boundary more than in other cases, leading to the formation of two small “kinks” (or nearly closed loops) near the sink. These kinks can be seen in Figs. 9, 12, 14, and 15. The kinks do not appear to significantly influence fluid flow, but they suggest that increasing values of the spring coefficient K_B may be needed as the spatial grid is refined.

5.2. Temporal Convergence Study

Temporal convergence studies were conducted for the pressure and velocity over the entire computational grid. Using the most refined grid, where $h = h_o/4$, the initial model

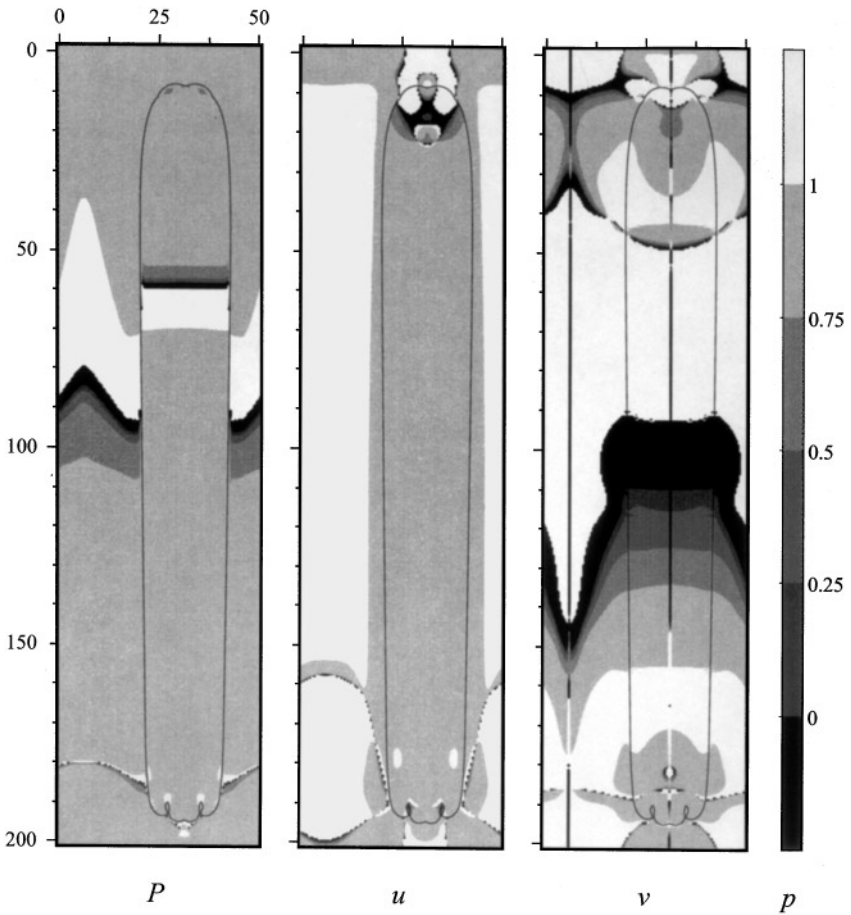


FIG. 9. Gray-scale shading represents temporal convergence for fluid pressure P , x -velocity u , and y -velocity v . Convergence is expressed as the order of convergence p at each point on the most refined spatial grid ($h_o/4$). The order calculation was based on three numerical trials (starting at $\approx 2/175$ s) with time steps $\Delta t/2^i$, $i = 0, 1, 2$. For P , u , and v , monotonicity is observed at 99.3%, 99.6%, and 93.3% of the spatial grid points, respectively. The average order of convergence for monotone regions is 0.896, 0.971, and 1.10 for P , u , and v , respectively. Black regions correspond to nonmonotonic values. Solid line indicates the vessel wall. Length scales are in micrometers.

arteriole configuration was pressurized ($\bar{P}_1 = 100$ mm Hg, $\bar{P}_2 = 0$ mm Hg) for $2/175$ s (with $\Delta t = 5/175 \times 10^{-5}$ s). Starting from $2/175$ s, the simulation was continued with a time step of Δt for 10 iterations, with a time step of $\Delta t/2$ for 20 iterations, and with a time step of $\Delta t/4$ for 40 iterations. The order of convergence p in time at each point on the computational grid was determined for the fluid pressure P , x -velocity u , and y -velocity v , using the results of these three continued simulations. Results are shown in Fig. 9. If the three values at a particular grid point were not monotonic, the corresponding grid-point region was shaded black. The average values of order of convergence, based only on monotone regions, were 0.896, 0.971, and 1.10, corresponding to P , u , and v , respectively.

5.3. Spatial Convergence Studies

Spatial convergence studies were conducted for fluid pressure and velocity over the entire computational grid. Figure 10 shows the order of convergence p in space at each point on

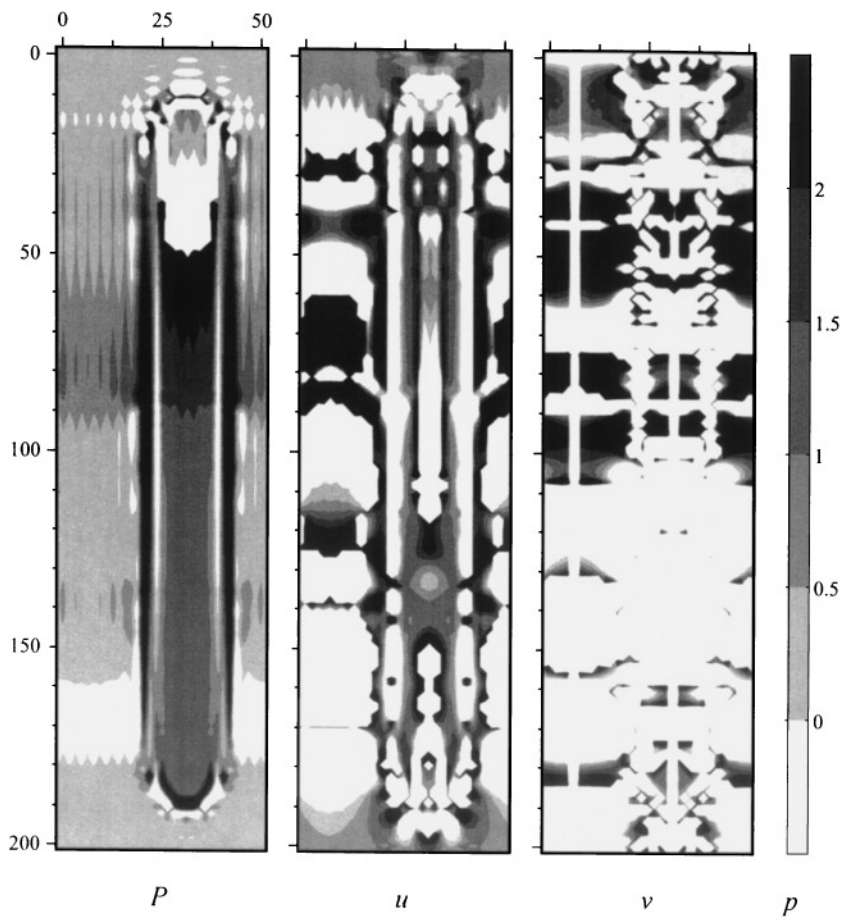


FIG. 10. Gray-scale shading represents spatial convergence for fluid pressure P , x -velocity u , and y -velocity v . Convergence is expressed as the order of convergence p at each point on the coarse computational grid ($h = h_o$) for the quasi-steady-state configuration, based on the three levels of spatial grid refinement. For the pressure P and the x - and y -velocities (u and v), monotonicity is observed at 96.9%, 70.4%, and 53.1% of the spatial grid points corresponding to the grid points of the coarsest grid. In this figure (unlike Fig. 9), white regions correspond to nonmonotonic sites. Convergence at monotonic sites is good or excellent throughout most of the domain; the average order of convergence is 0.721, 1.37, and 1.21 for P , u , and v , respectively. Length scales are in micrometers.

the coarse computational grid ($h = h_o$), based on the three levels of spatial grid refinement at an elapsed time of ~ 0.07 s, which corresponds to quasi-steady-state flow. Average order of convergence, for monotone regions, is 0.721, 1.37, and 1.21 for p , u , and v , respectively.

Although the computed value for the order of convergence for the pressure is substantially less than the order of convergence for the velocities, monotonicity is observed at 96.9% of the pressure values, while monotonicity is preserved at only 70.4% of the u values and 53.1% of the v values. Lack of monotonicity is especially prevalent near the boundary, as can be seen in Fig. 10. The lack of monotonic convergence near the boundary is expected, since the discretized boundary varies with spatial refinement: As the grid is refined, the approximate delta functions forming the vessel wall decrease in diameter. So not only is the fluid grid being refined, but the width available for fluid flow within the model arteriole is substantially increasing—by about 14% as h varies from h_o to $h_o/4$. Thus, the effective fluid domain is changing size as the spatial grid is refined.

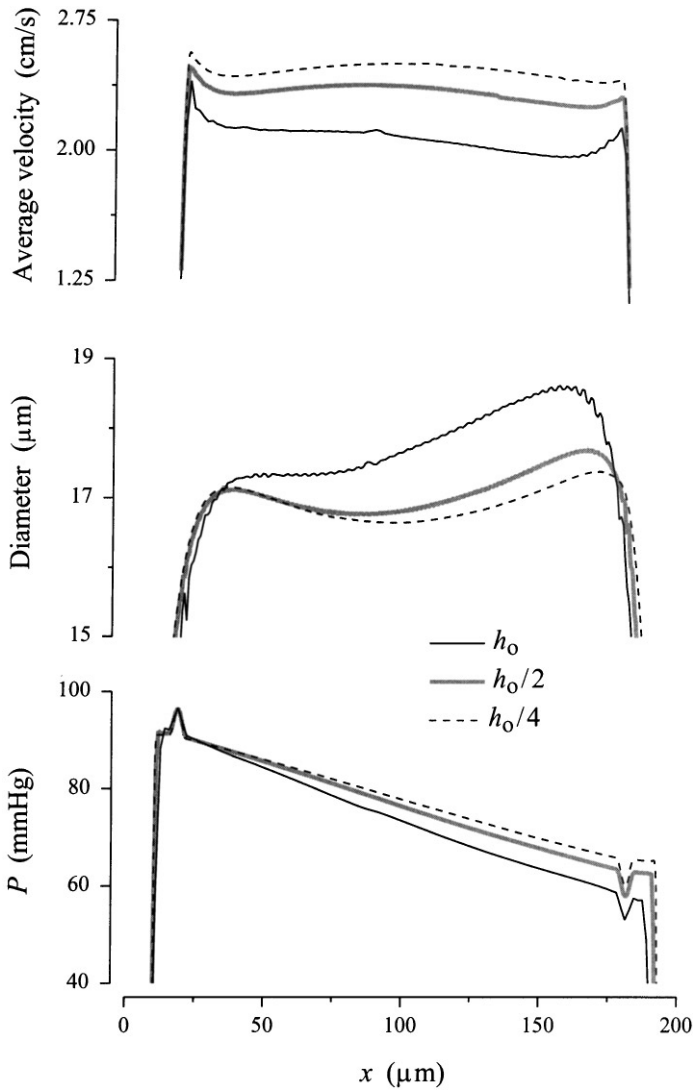


FIG. 11. Average x -velocity, vessel diameter, and pressure along the length of the model arteriole for the three levels of spatial grid refinement. For each variable, good convergence along most of the length of the model arteriole can be seen; the average order of convergence between 60 and 160 μm is 1.187 for the average velocity, 2.258 for the diameter and 1.117 for the pressure. Oscillations in diameter and velocity are substantially reduced by grid refinement. Source and sink locations are distinguishable as local deviations on the pressure profiles.

A comparison of the average velocity, diameter, and fluid pressure along the length of the model arteriole for the three spatial grid refinements is given in Fig. 11. Good convergence can be seen along most of the model arteriole in each case; the average order of convergence between 60 μm and 160 μm is 1.187 for the average velocity, 2.258 for the diameter, and 1.117 for the pressure. Oscillations in the diameter and average velocity are substantially reduced by grid refinement.

Previous convergence studies of the IMBM have used different convergence criteria: The order of convergence was evaluated by computing the ratios of the relative differences of velocities at successive levels of refinement, with respect to the ℓ_1 , ℓ_2 , and ℓ_∞

TABLE IV
Convergence Study of Fluid Velocity and Pressure

Convergence measure	Norm		
	ℓ_1	ℓ_2	ℓ_∞
$A = \frac{\ \mathbf{u}_{h_o/4} - \mathbf{u}_{h_o/2}\ }{\ \mathbf{u}_{h_o/4}\ }$	0.07372	0.05697	0.07018
$B = \frac{\ \mathbf{u}_{h_o/4} - \mathbf{u}_{h_o}\ }{\ \mathbf{u}_{h_o/4}\ }$	0.20901	0.15651	0.22732
Convergence ratio (A/B)	0.35271	0.36400	0.30873
Order of convergence p for $\mathbf{u}$	0.87593	0.80509	1.16290
$C = \frac{\ P_{h_o/4} - P_{h_o/2}\ }{\ P_{h_o/4}\ }$	0.04754	0.10992	0.68464
$D = \frac{\ P_{h_o/4} - P_{h_o}\ }{\ P_{h_o/4}\ }$	0.12943	0.24217	1.04328
Convergence ratio (C/D)	0.36730	0.45390	0.65624
Order of convergence p for P	0.78456	0.26679	-0.93282 ^a

Note. The convergence ratios reported here were determined using the specified norm over all computational points for the spatial grid refinement $h = h_o$. The order of convergence was determined using Eq. (A2), where q is the convergence ratio.

^a The negative value indicates that pressure does not converge in the ℓ_∞ norm.

norms, evaluated over the *entire* fluid domain [10, 28]. These studies demonstrated first-order spatial convergence. We have used these criteria to evaluate our simulation results, for quasi-steady-state, using the velocity on the most refined grid ($h_o/4$) as the reference, i.e., $(\|\mathbf{u}_{h_o/4} - \mathbf{u}_h\|_p)/\|\mathbf{u}_{h_o/4}\|_p$, where $h = h_o, h_o/2$, and $p = 1, 2, \infty$. Table IV shows that the convergence results from our simulations, using these criteria, are consistent with first-order convergence. Because fluid pressure is used in the determination of model arteriole diameter, Table IV also includes results for pressure convergence; previous convergence studies did not include fluid pressure, since it was considered an intermediate step in calculating velocity.

Thus far, measures of spatial convergence have been applied to fluid variables in the domain. Since we are also seeking to simulate the motion of vasoactive substances in the arteriolar flow, it is also important to measure the spatial convergence of solute that is advected with the local fluid velocity. To test the spatial convergence of solute distribution, a solute bolus was introduced into quasi-steady-state flow, and the resulting solute advection was observed. Initially, the solute concentration, measured in units of micromolar (μM), was a Gaussian centered at the coordinates (39.0625 μm , 31.2500 μm) and given by

$$c(x, y, 0) = \exp\left(\frac{-(x - 39.0625)^2}{2(2.01875)^2}\right) \exp\left(\frac{-(y - 31.2500)^2}{2(2.01875)^2}\right). \tag{71}$$

A solute advection comparison for the three levels of spatial grid refinement is shown in Fig. 12: h_o , panels A1–A4; $h_o/2$, panels B1–B4; $h_o/4$, panels C1–C4. The first column corresponds to an elapsed time of 3/3500 s after the bolus was introduced; each successive column is time-advanced by 3/3500 s. Truncation error significantly affects solute conservation; indeed, after an elapsed time of 9/3500 s, 93.3% of the initial solute remains for h_o , 98.6% remains for $h_o/2$, and 99.2% remains for $h_o/4$. Concentration profiles along the line $y = 31.25 \mu\text{m}$ are shown in Fig. 13 for each of the times shown in Fig. 12. The

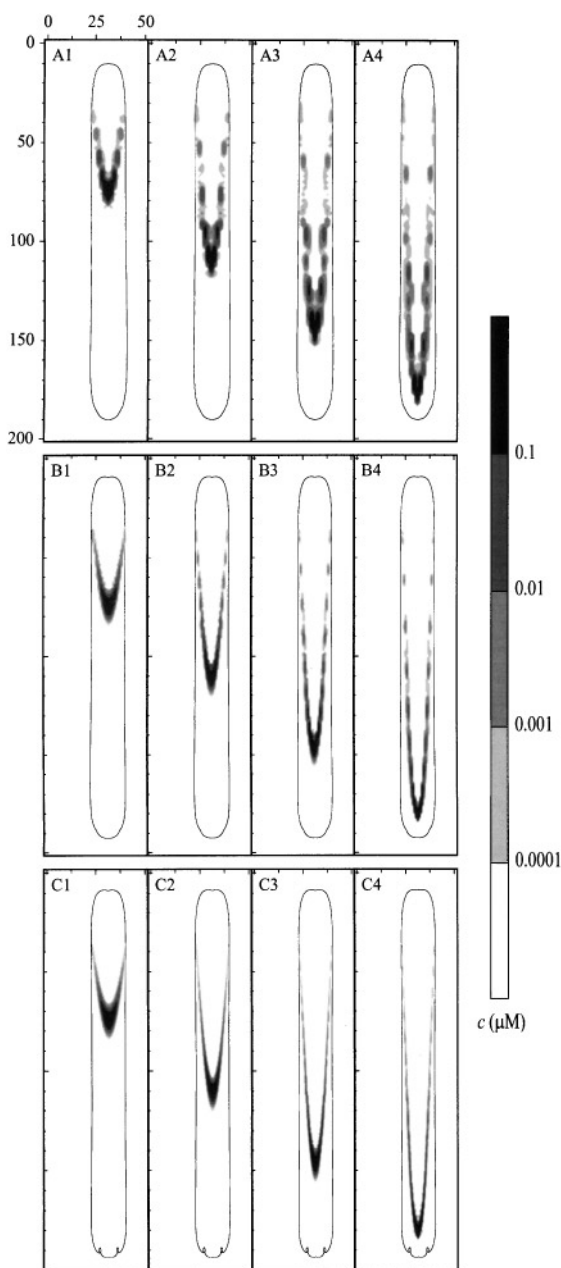


FIG. 12. Advection of solute bolus in quasi-steady-state flow for the three levels of spatial grid refinement: h_o , panels A1–A4; $h_o/2$, panels B1–B4; $h_o/4$, panels C1–C4. The first column corresponds to dimensional time $t = 3/3500$ s; each successive column is time-advanced by $3/3500$ s. The bar at the right indicates the concentration scale in micromolar. Initially the bolus has a Gaussian distribution. The figure shows that the eighth-order method employed controls artifactual diffusion, but that substantial grid refinement is needed for accurate representation of solute distribution in time. For the simulations shown in column 4, solute has begun to leave the domain through the fluid sink. The solid line indicates the vessel wall. Length scales are in micrometers.

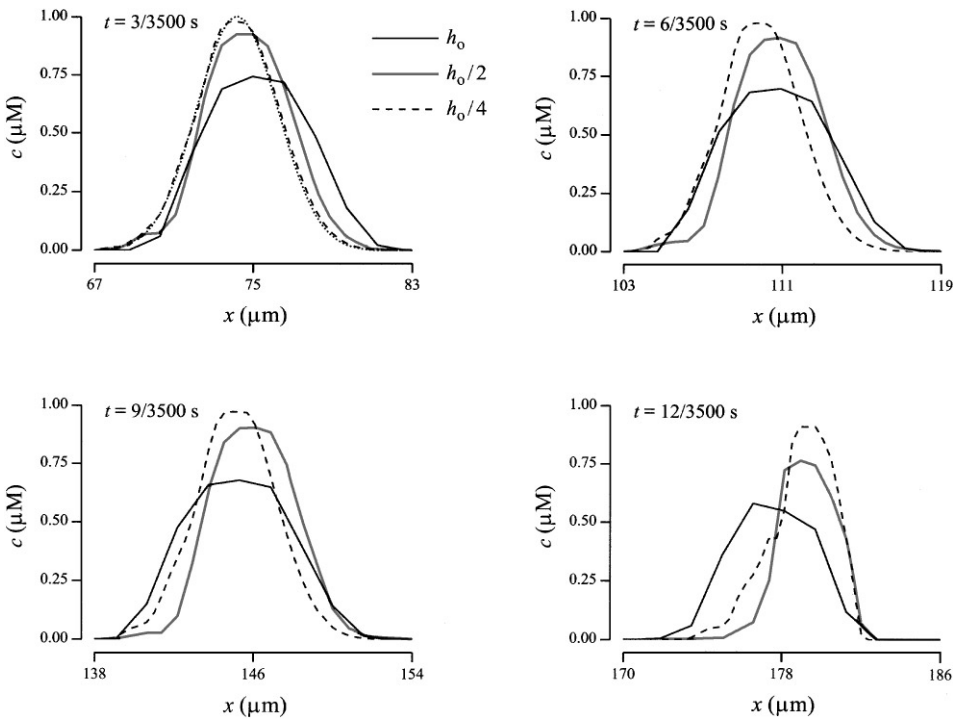


FIG. 13. Concentration profiles along the vertical line of symmetry through the model arteriole (i.e., along $y = 31.25 \mu\text{m}$) for the cases shown in Fig. 12. The elapsed time is labeled on each panel, corresponding to columns 1–4 in Fig. 12. The dotted line in panel 1 shows the initial solute profile translated by the distance that the maximum solute concentration advected for the most refined spatial grid. Solute has begun leaving the domain in panel 4, which explains the clipped shape of the leading edge of the curves. The decrease in maximum concentration with decreasing grid refinement can be seen; the maximum concentration for each profile is given in Table V. The figure also shows that the velocity along $y = 31.25 \mu\text{m}$ varies with each spatial grid refinement.

maximum concentration of the bolus along the line $y = 31.25 \mu\text{m}$ for each level of spatial grid refinement is reported in Table V. From this comparison, we conclude that a minimum grid refinement of $h_o/2$ is required to acceptably simulate solute advection within the model arteriole.

In the final simulation, NO was released from opposing locations along the model arteriole wall. The NO advects with the local fluid velocity, diffuses, degrades, and, as a function of local concentration, dilates the vessel diameter. Based on Eq. (67) NO was released from

TABLE V
Maximum Concentration (μM) from Figs. 12 and 13

Elapsed time (s)	h_o	$h_o/2$	$h_o/4$
0	1.000000	1.000000	1.000000
3/3500	0.740285	0.920517	0.975542
6/3500	0.693732	0.906282	0.973109
9/3500	0.678352	0.904106	0.972825

Note. Concentrations are not provided for the elapsed time 12/3500 s because at that time a significant amount of solute has exited the domain at the fluid sink.

TABLE VI

Distance (μm) along Model Arteriolar Wall from NO Source That Corresponds to Specified NO Concentrations for Each Level of Spatial Grid Refinement Shown in Fig. 14

Elapsed time (s)	1 μM			0.1 μM			0.01 μM		
	h_o	$h_o/2$	$h_o/4$	h_o	$h_o/2$	$h_o/4$	h_o	$h_o/2$	$h_o/4$
1/700	3.87	4.53	4.83	11.56	12.49	13.43	17.66	20.10	21.90
2/700	6.02	6.82	7.01	18.35	21.42	23.85	34.16	41.87	45.52
3/700	6.92	8.23	8.36	24.70	30.71	34.26	54.38	68.93	74.57
4/700	7.72	9.15	9.24	28.85	37.37	43.40	79.10	96.52	104.40

the wall at the rate of $8.32 \text{ nmol}/(\text{cm}^3 \cdot \text{s})$ from an approximate delta function of side $4h_o$ centered at the boundary nodes whose unstressed locations are given (in micrometers) by the coordinates (56.2500, 26.5625) and (56.2500, 35.9375). Results from the simulation for the three levels of spatial grid refinement are shown in Fig. 14. The first column corresponds to an elapsed time of 1/700 s after NO was introduced into the quasi-steady-state configuration; successive columns are time-advanced by 1/700 s. Table VI provides the distance in micrometers, measured along the model arteriole wall relative to the NO source, where concentration levels of 0.1, 0.01, and $0.001 \mu\text{M}$ are found. There are significant differences between the distances NO is advected for the levels of spatial grid refinement, especially for the first refinement to $h_o/2$, owing in part to the thinner boundary and the more accurate fluid velocities computed near the boundary.

In this simulation, the periodic domain used *for solute diffusion and degradation only* was twice as large in the y-coordinate as the fluid domain. This increase in domain size was needed to track the significant distances solute can advance owing to diffusion. Since fluid motion outside of the model arteriole was essentially zero, the larger domain had no effect on solute advection. In panel A of Fig. 15, at an elapsed time of 6/700 s, the effects of solute wrap-around owing to diffusion can be seen. After 12/700 s, solute has almost completely filled the domain as shown in panel B. In future simulations, a larger solute domain will be required, or perhaps a different methodology, such as absorbing boundary conditions, may be used [17].

Figure 16 shows the NO concentration along the model wall and the diameter and fluid pressure along the length of the model arteriole at a grid refinement of $h = h_o/4$ for the quasi-steady-state configuration (dashed line) and for the simulation arterioles shown in Figs. 14 and 15. Significant vessel dilation is seen with increasing time, with the final time of 12/700 s (gray line) corresponding to an almost completely relaxed arteriole. Thus, although NO is being removed at a rapid rate, these simulations show that there is sufficient velocity near the walls to advect NO the entire length of the model arteriole.

6. DISCUSSION

We have developed a model of arteriolar fluid flow that includes a phenomenological model of the myogenic response of the arteriolar wall and that can include the release, distribution, and action of vasoactive substances. We have used the model to simulate hemodynamic regulation in the renal afferent arteriole, where the myogenic response and the vasodilatory action of NO are components of a complex system that regulates the rate

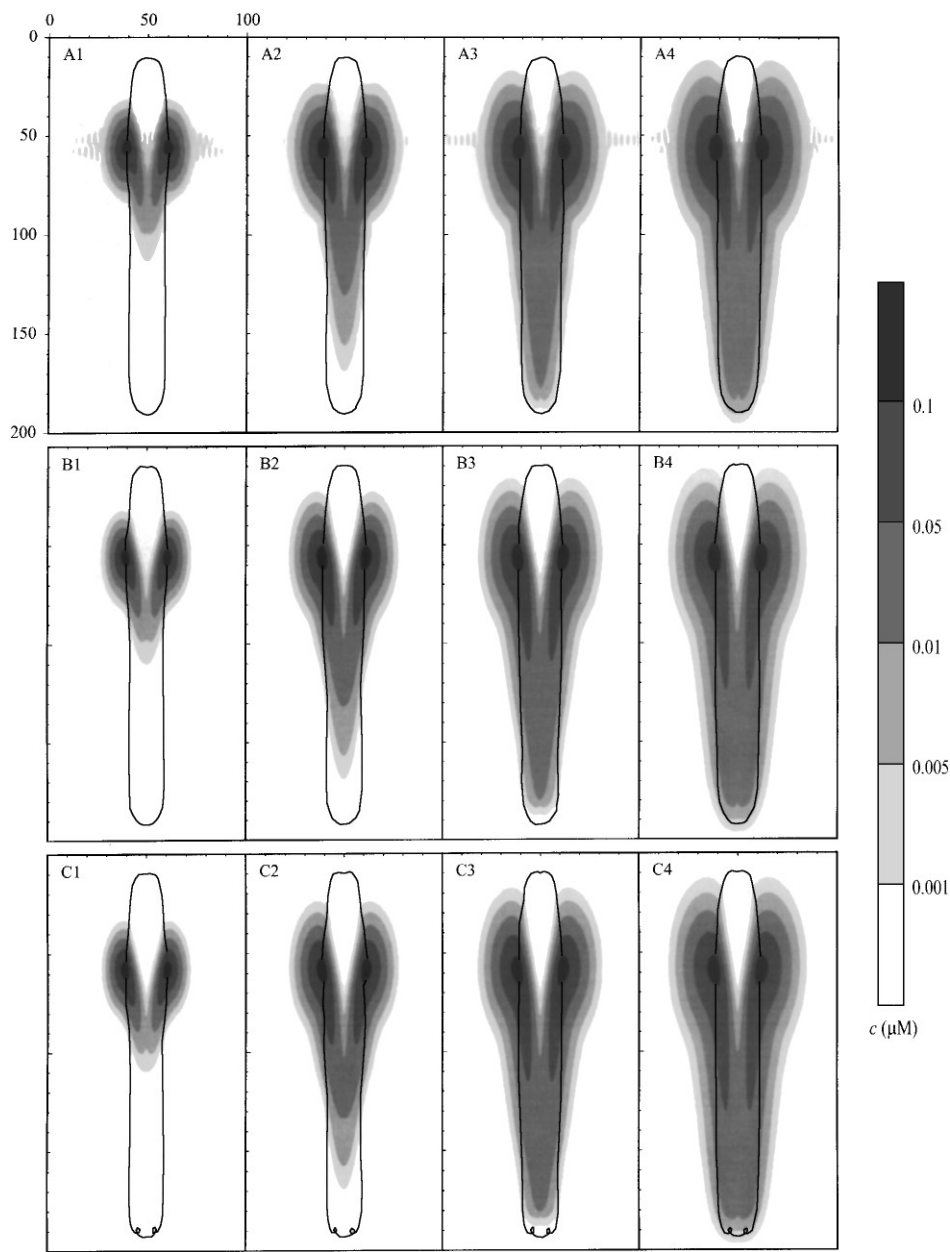


FIG. 14. Release, advection, diffusion, and degradation of NO for the three levels of spatial grid refinement: h_o , panels A1–A4; $h_o/2$, panels B1–B4; $h_o/4$, panels C1–C4. The first column corresponds to dimensional time $t = 1/700$ s after NO was introduced into established quasi-steady-state flow; each successive column is time-advanced by $1/700$ s. The bar at the right indicates the NO concentration scale in micromolar. NO is released from two sites fixed on the model arteriolar wall. Table VI provides the distance along the arteriolar wall from the NO source that corresponds to NO concentrations of 1.0, 0.1, and $0.01 \mu\text{M}$ for each simulation shown. The solid line indicates the vessel wall. Length scales are in micrometers. To better represent NO diffusion, the spatial grid used for diffusion was constructed to be twice as wide as that used for the fluid domain (see a more detailed explanation in the text).

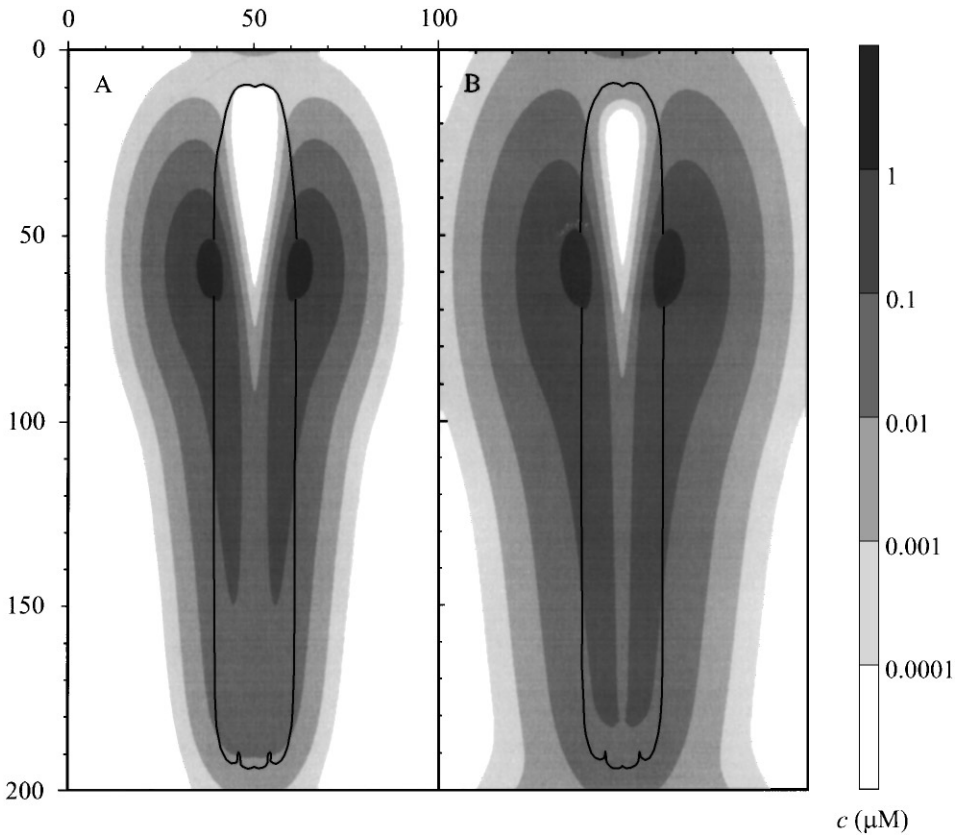


FIG. 15. NO concentration on the grid refinement $h = h_o/4$ after longer elapsed times than those shown in Fig. 14. Panel A shows an elapsed time of $6/700$ s; the solute that has diffused to the bottom of the domain begins to wrap around the periodic domain and to overlap at the top, owing to the periodic boundary. Panel B illustrates an elapsed time of $12/700$ s; the solute diffused at the sides begins to wrap around the domain. The solid line indicates the vessel wall. Length scales are in micrometers.

of fluid entry into the nephrons of the kidney. This regulation helps maintain the balance between the reabsorptive capacity of the nephron and the fluid load. Moreover, the regulatory action of the afferent arteriole protects the delicate glomerular capillaries from injury.

Numerical solutions to the model fluid equations were obtained by means of the IMBM. Advection of NO was computed by means of a high-order flux-corrected transport scheme, and diffusion of NO was computed by a spectral solver. We investigated the temporal and spatial convergence properties of model variables, especially near the boundary; these convergence studies were made possible, in part, by the simple model geometry, which produces an approximation to planar Poiseuille flow. As shown in Section 5.3, a spatial grid refinement with $h < 1 \mu\text{m}$ is required to track fluid and solute with sufficient accuracy for physiologically meaningful simulations. This grid refinement corresponds to approximately $1/25$ th the diameter of the model arteriole. The methods developed and tested in this study could be applied to other physiological tubular structures that have low Reynolds number flow, e.g., to other small vessels, to vessel networks, or to the tubules of the nephron.

The numerical methods utilized in this study are formulated with first-order differencing in time, and the convergence study of Section 5.2 indicates that first-order temporal

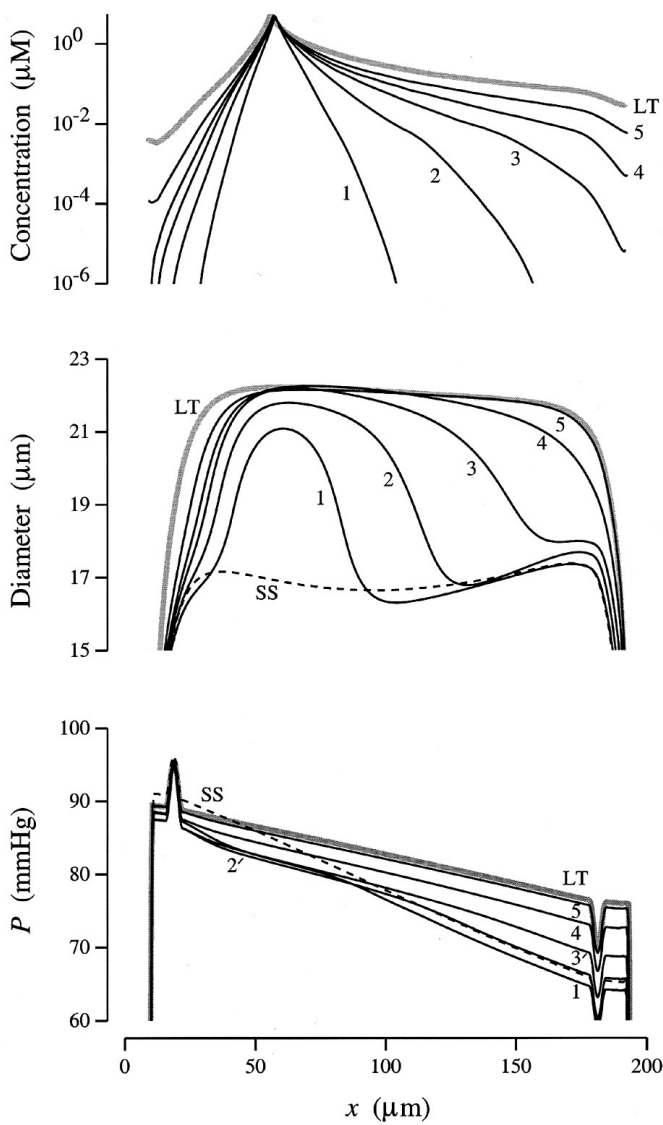


FIG. 16. The concentration along the model wall and the diameter and pressure along the length of the model arteriole are shown at a spatial grid refinement $h = h_o/4$ for the quasi-steady-state configuration and for the configurations shown in Figs. 14 and 15. Data for the model arteriole at elapsed times of 1/700, 2/700, 3/700, 4/700, 6/700, and 12/700 s, are labeled 1, 2, 3, 4, 5, and LT (gray line), respectively. Quasi-steady-state configuration is labeled SS (dashed line). (No curve labeled SS appears in the topmost panel, since the NO concentration is zero for the quasi-steady-state configuration.) Wall relaxation, owing to significant NO concentration, extends along the length of the model arteriole. After 12/700 s (LT), the arteriole has reached maximal relaxation for this level of NO.

convergence is attained over most of the fluid domain. Because very small time steps are required to maintain stability in the IMBM, first-order time differencing should provide a high degree of temporal accuracy.

Although our empirical evaluation indicates second-order spatial convergence in key quantities (see Sections 5.1 and 5.3), the spatial convergence properties of the numerical

methods differ. The tripartite splitting in the fluid solver uses first-order upwind differencing for the inertial (hyperbolic) term, a spectral method—which is high-order—for the viscous (parabolic) term, and a method of unknown spatial convergence for the force arising from the boundary nodes. The hyperbolic term, because its relative contribution to the solution is small, probably has little influence on the spatial convergence of the fluid solver. Indeed, in the calculation of the x -velocity u at the location $(100\ \mu\text{m}, 31.25\ \mu\text{m})$, the center of fully developed simulated arteriolar flow, the ratio of the viscous term magnitude to the inertial term magnitude is $\sim 10^5$.

An important scientific conclusion of this study is that NO transported in the intravascular flow can stimulate significant downstream vasodilation, because NO advection, relative to diffusion and degradation, is the dominating effect. This conclusion is consistent with the results of Lancaster whose one-dimensional model included NO diffusion, degradation, and binding with luminal hemoglobin [22]. Also, since experimental observations show that the myogenic response maintains a nearly constant vessel diameter as length along the vessel increases (and thus intravascular pressure decreases), this study suggests that there is an intrinsic mechanism that modulates the myogenic response so that it acts more strongly in the proximal arteriole, where the pressure is higher, than in the distal arteriole, where the pressure is lower.

Several computational issues merit more attention. As noted in Section 4, the time scale of the dynamic myogenic response was increased by more than two orders of magnitude so that the myogenic response could be simulated with currently available supercomputing resources. Depending on the phenomenon under investigation, this may not be a serious shortfall of the simulation. For example, in the current application, the distribution of NO is advection dominated, and the fluid transit time of the afferent arteriole is much faster than the characteristic time of the myogenic response. Indeed, the physiological myogenic response requires ~ 10 s, the accelerated myogenic response requires ~ 0.06 s, and the fluid transit time is ~ 0.006 s. To perform simulations on a true time scale, longer time steps would need to be taken, perhaps by using a more stable numerical method for incorporating the boundary force density.

In long-time simulations, the accumulation of machine error causes a loss of symmetry in the configuration of the model vessel wall, which may eventually lead to a loss of fluid containment. In Section 5 we described a numerical adjustment that acts to reduce the accumulation of machine roundoff error by adjusting the forces arising from the model of the myogenic response. However, a better method for controlling this error is needed for reliable long-time solutions. For more quantitatively accurate simulations, the model should be formulated in three-dimensional space, perhaps by using a fluid solver formulated in cylindrical coordinates, which would enforce radial symmetry.

Several physiological issues also merit attention. The diameter of red blood cells is about 0.4 of the diameter of the arteriole; blood cells are known to be concentrated near the center of the arteriolar flow and to therefore exert a spatially inhomogeneous effect on the flow characteristics [34]. Also, the NO carried by the plasma undergoes complicated chemical interactions with the hemoglobin in the red blood cells, including binding of NO to hemoglobin, saturation of binding sites, and downstream release of NO, where plasma O_2 concentration is reduced [35]. We have not modeled red blood cells (although this is possible by means of the IMBM), and we have not modeled the chemical interactions in this study. Moreover, several other important vasoactive substances (e.g., endothelin, a vasoconstrictor released by endothelial cells [26]) have been identified which could be included in a more

realistic model. An additional factor that influences afferent arteriolar muscle tension is the tubuloglomerular feedback (TGF) mechanism. Many aspects of TGF are well understood [33], and its effects could be integrated into the model.

Because our model for the myogenic response was based on experiments in which shear-stress-induced release of NO from endothelial cells was active, the model implicitly includes the response of the arteriolar wall to baseline levels of NO. In subsequent simulations, however, the myogenic response and arteriolar wall response to shear-stress-induced NO might be profitably decoupled. Such a decoupling could be important in the simulation of a focal constriction arising from injury [18].

In this study, we have not modeled the physical structure of the arteriolar wall, nor the mechanism of myogenic response; the IMBM is formulated as a boundary with singular support and our model for the myogenic response is phenomenological, i.e., it depends on the response to tubular pressure but does not incorporate the biochemical mediation and cellular generation of the contractile response. However, we have modeled the effect of the wall and the action of the myogenic mechanism in regulating arteriolar flow. The fluid interaction with the wall is of critical importance in determining the flow characteristics, and therefore, in this study, we conducted extensive tests to ascertain the degree of numerical refinement required for an accurate representation of that interaction. Because we intend to apply this model to realistic physiological contexts, this investigation, which serves to validate our methodology, is an essential first step.

APPENDIX

Here we provide explicit formulas for computing p , V_∞ , and K from Eq. (70). Let

$$q = \frac{V_2 - V_1}{V_2 - V_0}. \quad (\text{A1})$$

Then, after straightforward calculations, one finds

$$p = \frac{\ln(1/q - 1)}{\ln(2)}, \quad (\text{A2})$$

$$K = \frac{V_0 - V_1}{(h_o)^p(1 - 2^{-p})}, \quad (\text{A3})$$

and

$$V_\infty = V_0 - K(h_o)^p. \quad (\text{A4})$$

ACKNOWLEDGMENTS

This study is largely based on material that appeared in the first author's doctoral dissertation, *Flow Regulation in the Afferent Arteriole: An Application of the Immersed Boundary Method* (University Microfilms, Ann Arbor, MI, 1996). This research was supported in part by the National Institute of Diabetes and Digestive and Kidney Diseases through Grant DK42091 to H. E. Layton and the National Science Foundation through Group Infrastructure Grant DMS-9709608 to M. C. Reed, H. E. Layton, and J. J. Blum. Supercomputer time was provided by the North Carolina Supercomputing Center in the form of several allocations to H. E. Layton. We thank John M. Davies for assistance in figure preparation.

REFERENCES

1. P. L. Altman and D. S. Dittmer, *Respiration and Circulation* (Federation of American Societies for Experimental Biology, Bethesda, MD, 1971).
2. R. P. Beyer Jr., A computational model of the cochlea using the immersed boundary method, *J. Comput. Phys.* **98**, 145 (1992).
3. W. L. Briggs and V. E. Henson, *The DFT: An Owner's Manual for the Discrete Fourier Transform* (Soc. Indus. Appl. Math., Philadelphia, 1995).
4. R. Busse, I. Fleming, and V. B. Schini, Nitric oxide formation in the vascular wall: Regulation and functional implications, *Curr. Top. Microbiol. Immunol.* **196**, 7 (1995).
5. D. Casellas and L. C. Moore, Autoregulation of intravascular pressure in preglomerular juxtamedullary vessels, *Am. J. Physiol.* **264** [*Renal Fluid Electrolyte Physiol.* **33**], F315 (1993).
6. D. Casellas and L. C. Moore, Autoregulation and tubuloglomerular feedback in juxtamedullary glomerular arterioles, *Am. J. Physiol.* **258** [*Renal Fluid Electrolyte Physiol.* **27**], F660 (1990).
7. A. J. Chorin and J. E. Marsden, *A Mathematical Introduction to Fluid Mechanics* (Springer-Verlag, New York, 1993).
8. G. J. Christ, P. R. Brink, and S. V. Ramanan, Dynamic gap junctional communication: A delimiting model for tissue response, *Biophys. J.* **67**, 1335 (1994).
9. R. Dillon, L. Fauci, and D. Gaver III, A microscale model of bacterial swimming, chemotaxis, and substrate transport, *J. Theor. Biol.* **177**, 325 (1995).
10. R. Dillon, L. Fauci, A. Fogelson, and D. Gaver III, Modeling biofilm processes using the immersed boundary method, *J. Comput. Phys.* **129**, 57 (1996).
11. L. J. Fauci, Peristaltic pumping of solid particles, *Comp. Fluids* **21**, 583 (1992).
12. L. J. Fauci and A. McDonald, Sperm motility in the presence of boundaries, *Bull. Math. Biol.* **57**, 679 (1995).
13. L. J. Fauci and C. S. Peskin, A computational model of aquatic animal locomotion, *J. Comput. Phys.* **77**, 85 (1988).
14. A. L. Fogelson, A mathematical model and numerical method for studying platelet adhesion and aggregation during blood clotting, *J. Comput. Phys.* **56**, 111 (1984).
15. A. L. Fogelson and C. S. Peskin, A fast numerical method for solving the three-dimensional Stokes' equations in the presence of suspended particles, *J. Comput. Phys.* **79**, 50 (1988).
16. Y. C. Fung, *Biomechanics: Mechanical Properties of Living Tissues*, 2nd ed. (Springer-Verlag, New York, 1993).
17. L. Halpern and J. Rauch, Absorbing boundary conditions for diffusion equations, *Numer. Math.* **71**, 185 (1995).
18. G. Hansson, Y. Geng, J. Holm, P. Hårdhammar, Å. Wennmalm, and E. Jennische, Arterial smooth muscle cells express nitric oxide synthase in response to endothelial injury, *J. Exp. Med.* **180**, 733 (1994).
19. L. Jia, C. Bonaventura, J. Bonaventura, and J. S. Stamler, S-nitrosohaemoglobin: A dynamic activity of blood involved in vascular control, *Nature* **380**, 221 (1996).
20. C. T. Kelly, *Iterative Methods for Linear and Nonlinear Equations*, Frontiers in Applied Math., Vol. 16 (Soc. Indus. Appl. Math., Philadelphia, 1995).
21. M. Kelm, M. Feelisch, R. Spahr, H. Piper, E. Noack, and J. Schrader, Quantitative and kinetic characterization of nitric oxide and ERDF released from cultured endothelial cells, *Biochem. Biophys. Res. Commun.* **154**(1), 236 (1988).
22. J. R. Lancaster Jr., Simulation of the diffusion and reaction of endogenously produced nitric oxide, *Proc. Nat. Acad. Sci. USA* **91**, 8137 (1994).
23. H. H. Lipowsky and B. W. Zweifach, Methods for the simultaneous measurement of pressure differentials and flow in single unbranched vessels of the microcirculation for rheological studies, *Microvasc. Res.* **14**, 345 (1977).
24. T. F. Luescher, Imbalance of endothelium-derived relaxing and contracting factors: A new concept in hypertension, *Am. J. Hypertension* **3**, 317 (1990).

25. T. Malinski, Z. Taha, S. Grunfeld, S. Patton, M. Kapturczak, and P. Tombouliau, Diffusion of nitric oxide in the aorta wall monitored in situ by porphyrinic microsensors, *Biochem. Biophys. Res. Commun.* **193**, 1076 (1993).
26. B. O. Oluwole, M. McMillen, and B. E. Sumpio, Endothelial cell control of vasomotor tone, *Ann. Vasc. Surg.* **9**, 293 (1995).
27. C. S. Peskin, Numerical analysis of blood flow in the heart, *J. Comput. Phys.* **25**, 220 (1977).
28. C. S. Peskin and D. M. McQueen, A three-dimensional computational method for blood flow in the heart. I. Immersed elastic fibers in a viscous incompressible fluid, *J. Comput. Phys.* **81**, 372 (1989).
29. C. S. Peskin and D. M. McQueen, A general method for the computer simulation of biological systems interacting with fluids, in *Biological Fluid Dynamics*, edited by C. P. Ellington and T. J. Pedley, Symposia of the Society for Experimental Biology, No. 49 (Soc. Exp. Biol. Company of Biologists Ltd., Cambridge, 1995), p. 265.
30. W. H. Press, S. A. Teukolsky, W. T. Vetterling, and B. P. Flannery, *Numerical Recipes in Fortran: The Art of Scientific Computing*, 2d ed. (Cambridge Univ. Press, New York, 1992).
31. N. Rashevsky, *Mathematical Biophysics: Physico-Mathematical Foundations of Biology*, Vol. 1 (Dover, New York, 1960).
32. B. D. Rose and H. G. Rennke, *Renal Pathophysiology—The essentials* (Williams & Wilkins, Baltimore, 1994).
33. J. Schnermann and J. Briggs, Function of the juxtaglomerular apparatus: Control of glomerular hemodynamics and renin secretion, in *The Kidney: Physiology and Pathophysiology* (2d ed.), edited by D. W. Seldin and G. Giebisch (Raven, New York, 1992), p. 1249.
34. T. W. Secomb, Mechanics of blood flow in the microcirculation, in *Biological Fluid Dynamics*, edited by C. P. Ellington and T. J. Pedley, Symposia of the Society for Experimental Biology, No. 49 (Soc. Exp. Biol. Company of Biologists Ltd., Cambridge, 1995), p. 305.
35. J. S. Stamler, L. Jia, J. P. Eu, T. J. McMahon, I. T. Demchenko, J. Bonaventura, K. Gernert, and C. A. Piantadosi, Blood flow regulation by S-nitrosohemoglobin in the physiological oxygen gradient, *Science* **276**, 2034 (1997).
36. M. W. Vaughn, L. Kuo, and J. C. Liao, Modeling of endothelial nitric oxide production and diffusion, *FASEB J.* **11**(3), A288, Abstract No. 1673 (1997).
37. S. T. Zalesak, Fully multidimensional flux-corrected transport algorithms for fluids *J. Comput. Phys.* **31**, 335 (1979).
38. S. T. Zalesak, A preliminary comparison of modern shock-capturing schemes: Linear advection, in *Advances in Computer Methods for Partial Differential Equations VI: Proceedings of the 1987 IMACS Conference*, edited by R. Vichnevetsky and R. S. Stepleman (Internat. Assoc. Math. Comput. Simulation, New Brunswick, NJ, 1987), p. 15.